

Automated Removal of Acoustic Noise from fMRI-Acquired Jazz Scat Singing

Evan Szypulski

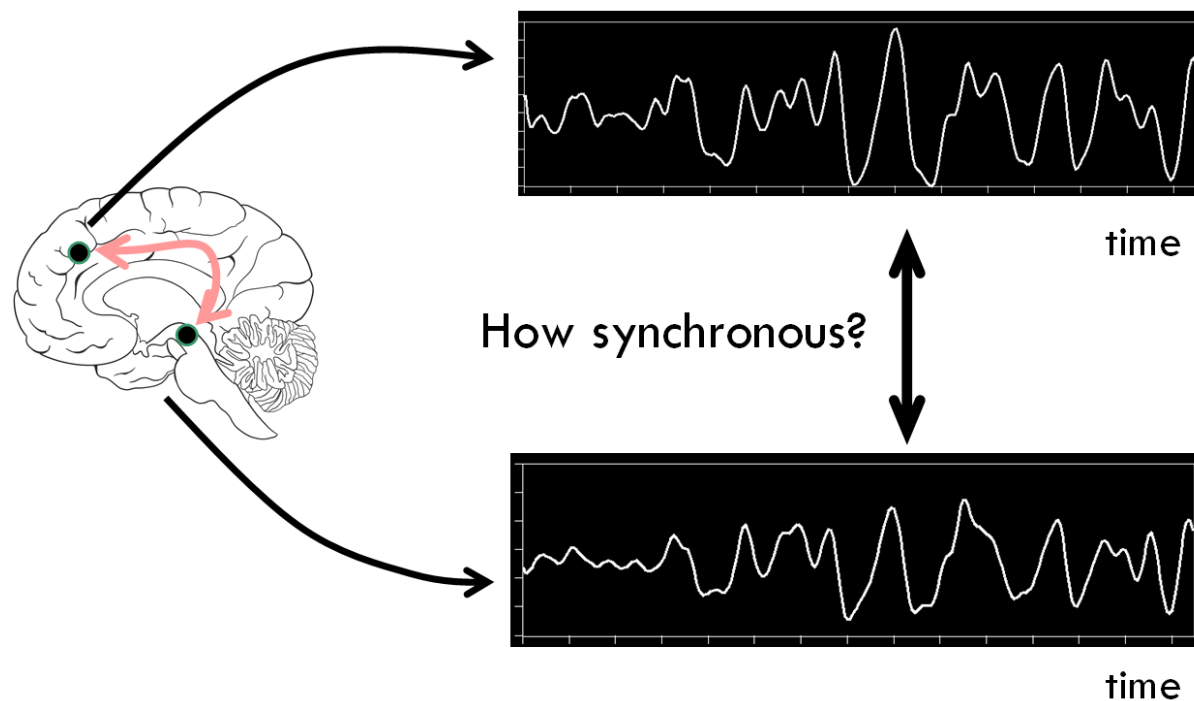
Project Sponsor: Dr. Martin Norgaard

GSURC 2026

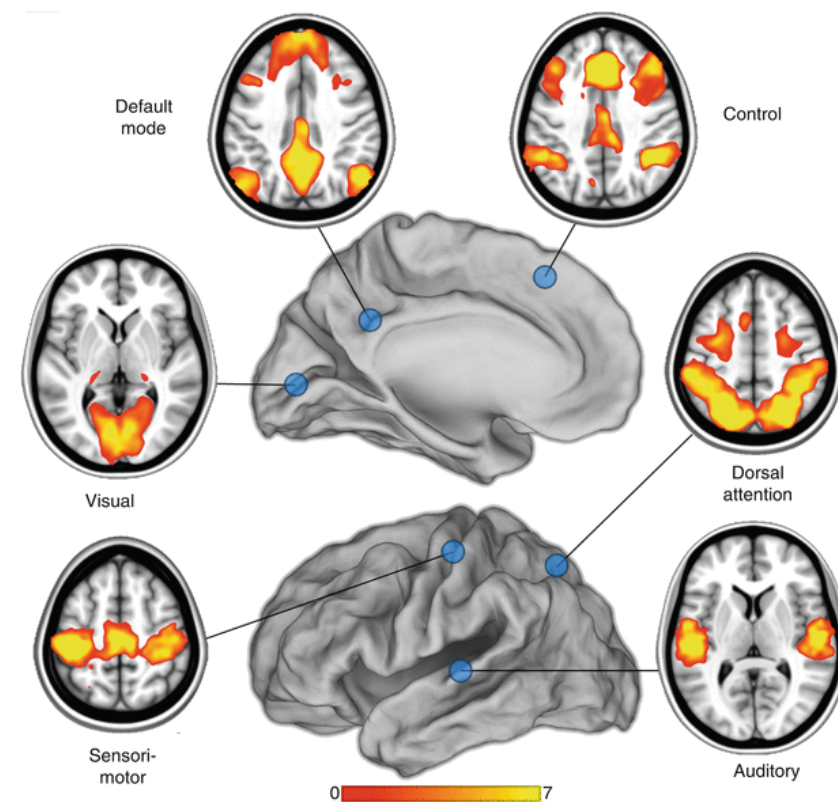


Functional network connectivity during Jazz improvisation

[Victor M. Vergara](#) , [Martin Norgaard](#), [Robyn Miller](#), [Roger E. Beaty](#), [Kiran Dhakal](#), [Mukesh Dhamala](#) & [Vince D. Calhoun](#)

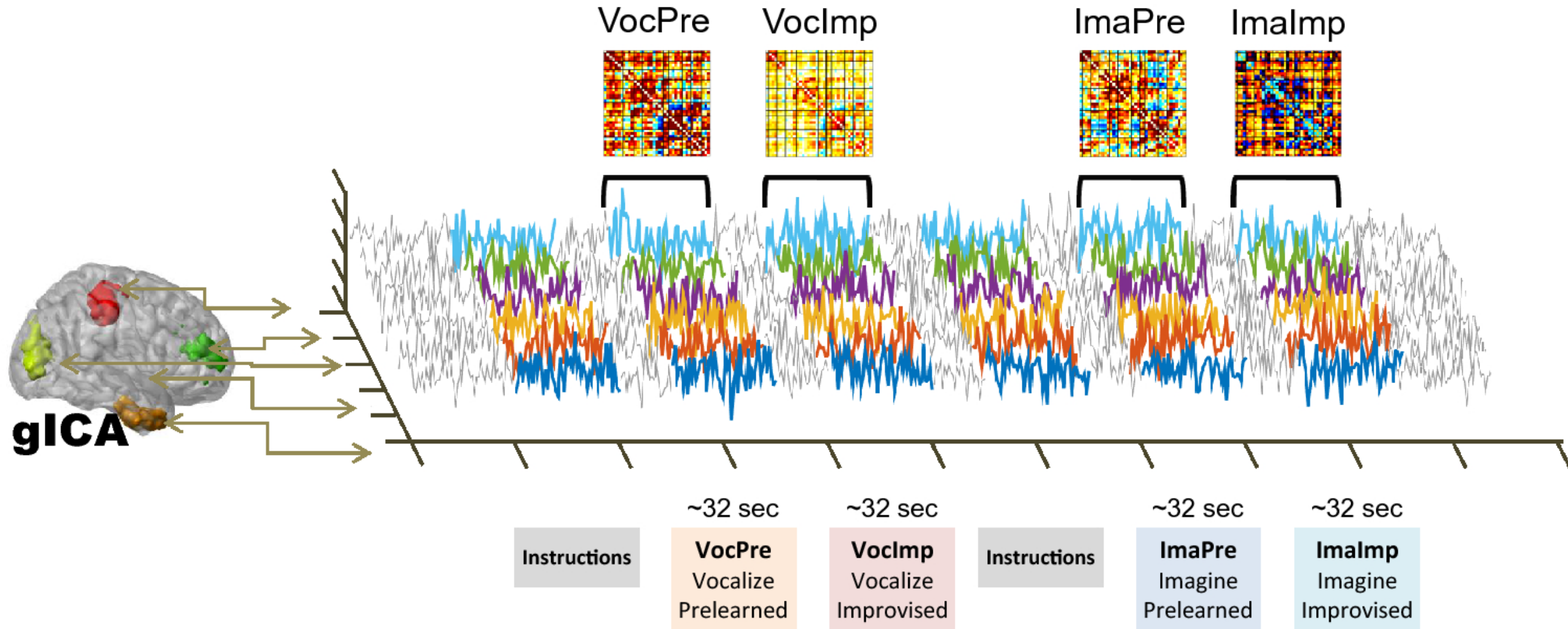


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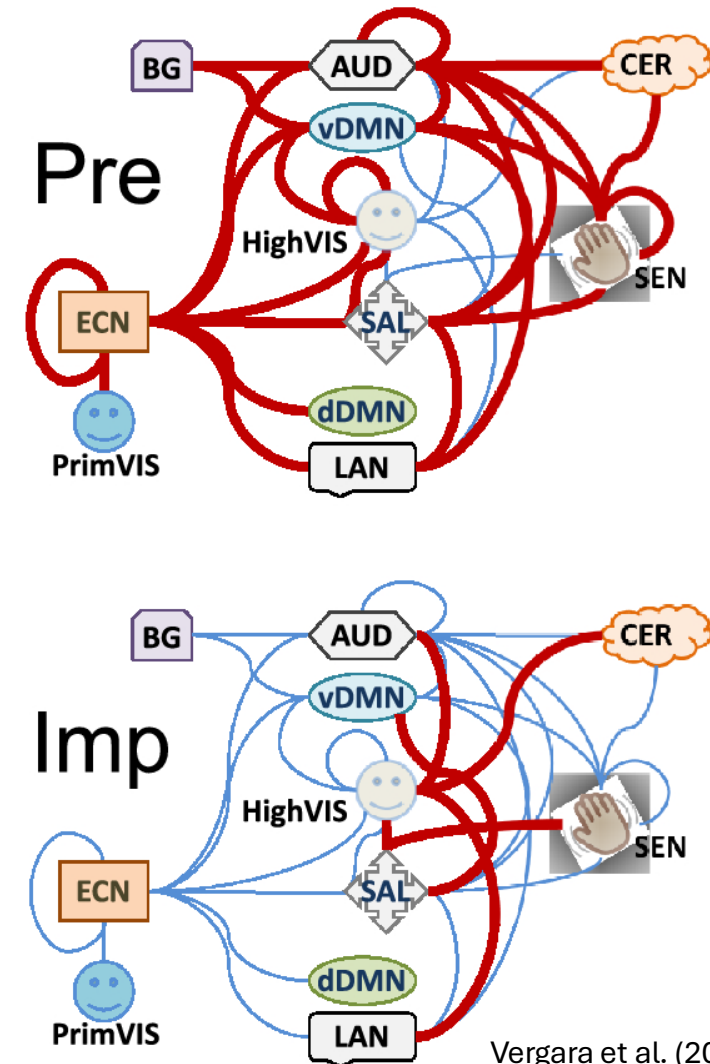
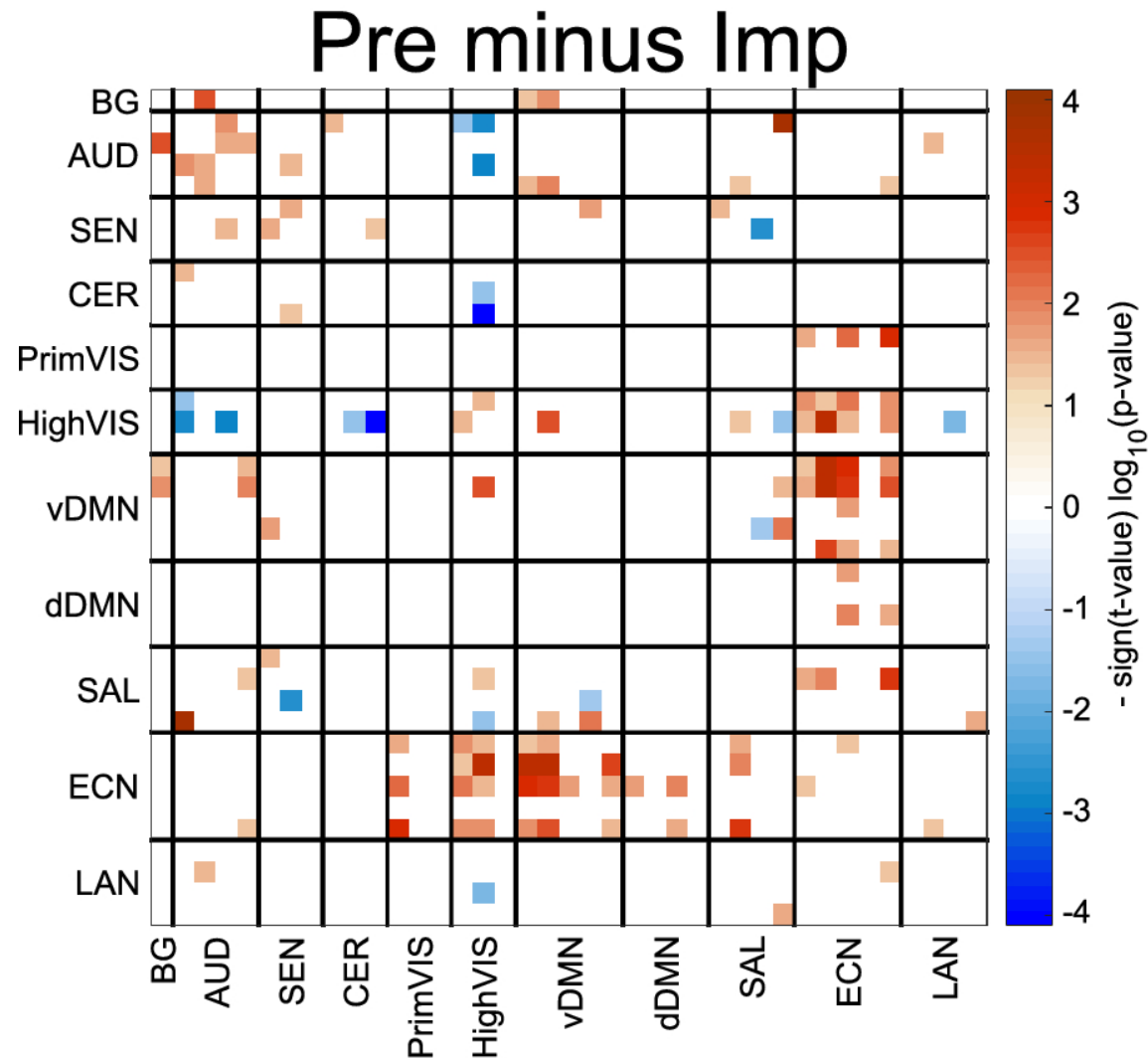


Zhang & Raichle (2010)

Each Subject Within Task Static-Functional-Network-Connectivity



Decreased ECN-DMN connectivity reflects “flow state”



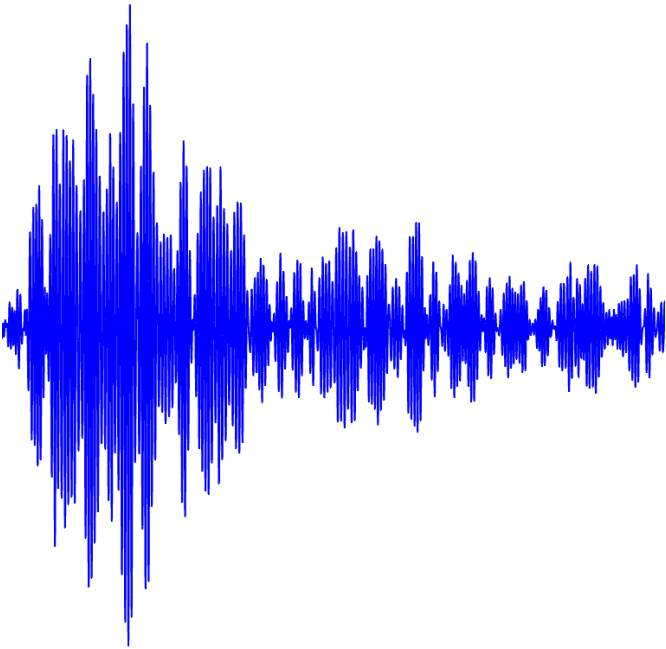
“Future research using the current vocalization task should capture high quality audio allowing for analysis of output.”

Vergara et al. (2021)

Scanner Audio

Expert Ratings

Brain Data



12:29

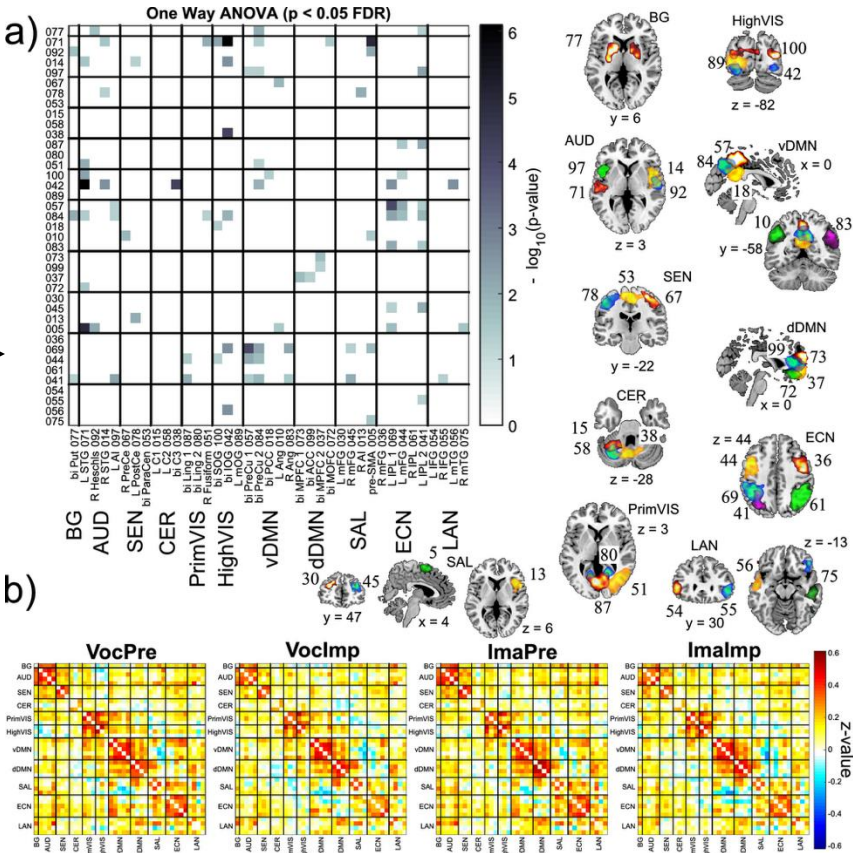
0:00 / 1:05

Instructions:
Please listen to the recording above.

- **0:00 ~ 0:15:** The soloist performs the end of the written melody.
- **~ 0:15 – End:** The soloist improvises over the harmonic progression.

After listening, please evaluate the improvisation section (0:15 onwards) using the criteria below.

How often does the soloist imply

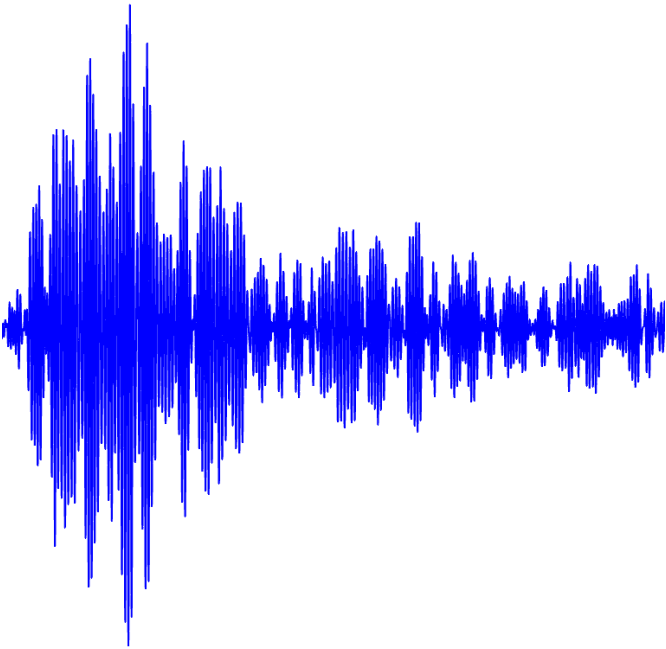


Vergara et al. (2021)

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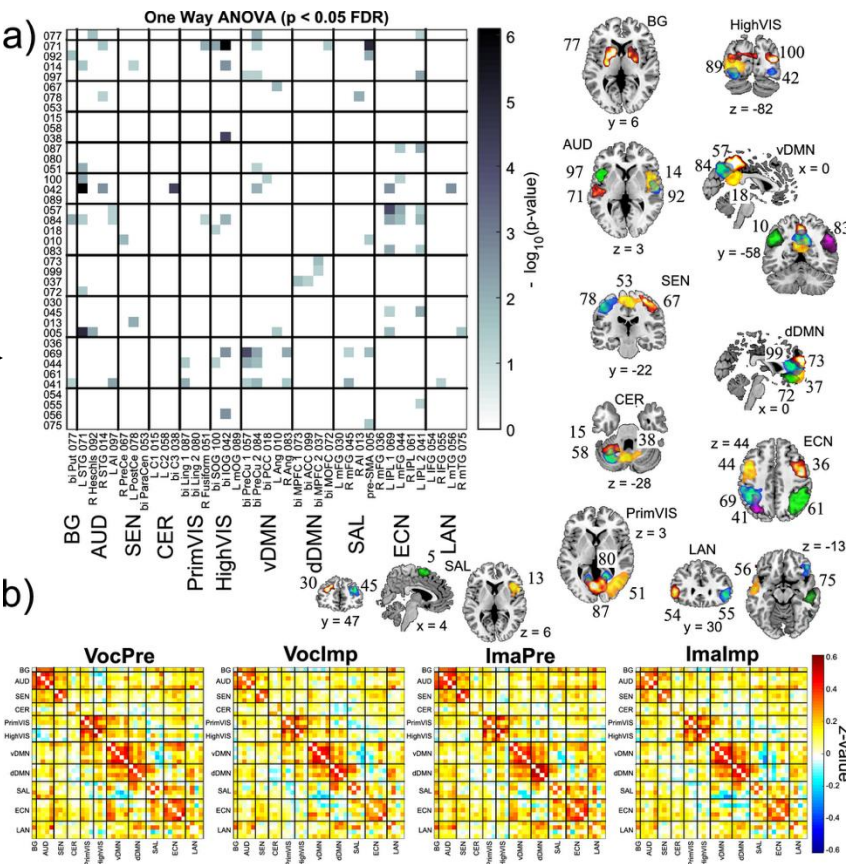
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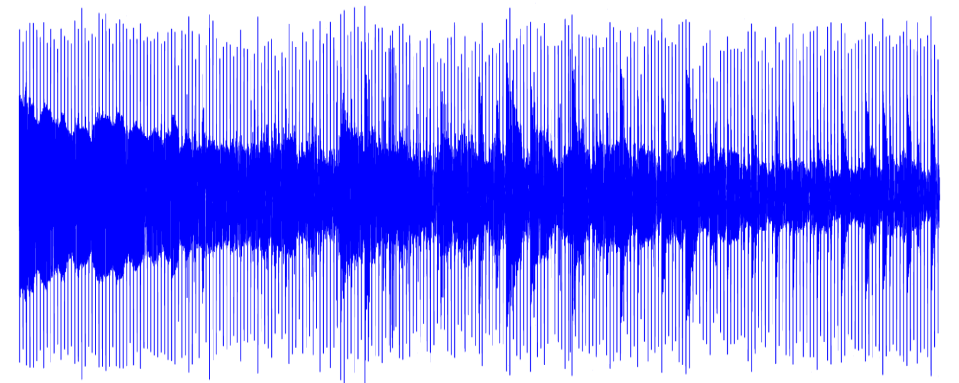
Vergara et al. (2021)

Simultaneous Audio and fMRI Data Acquisition

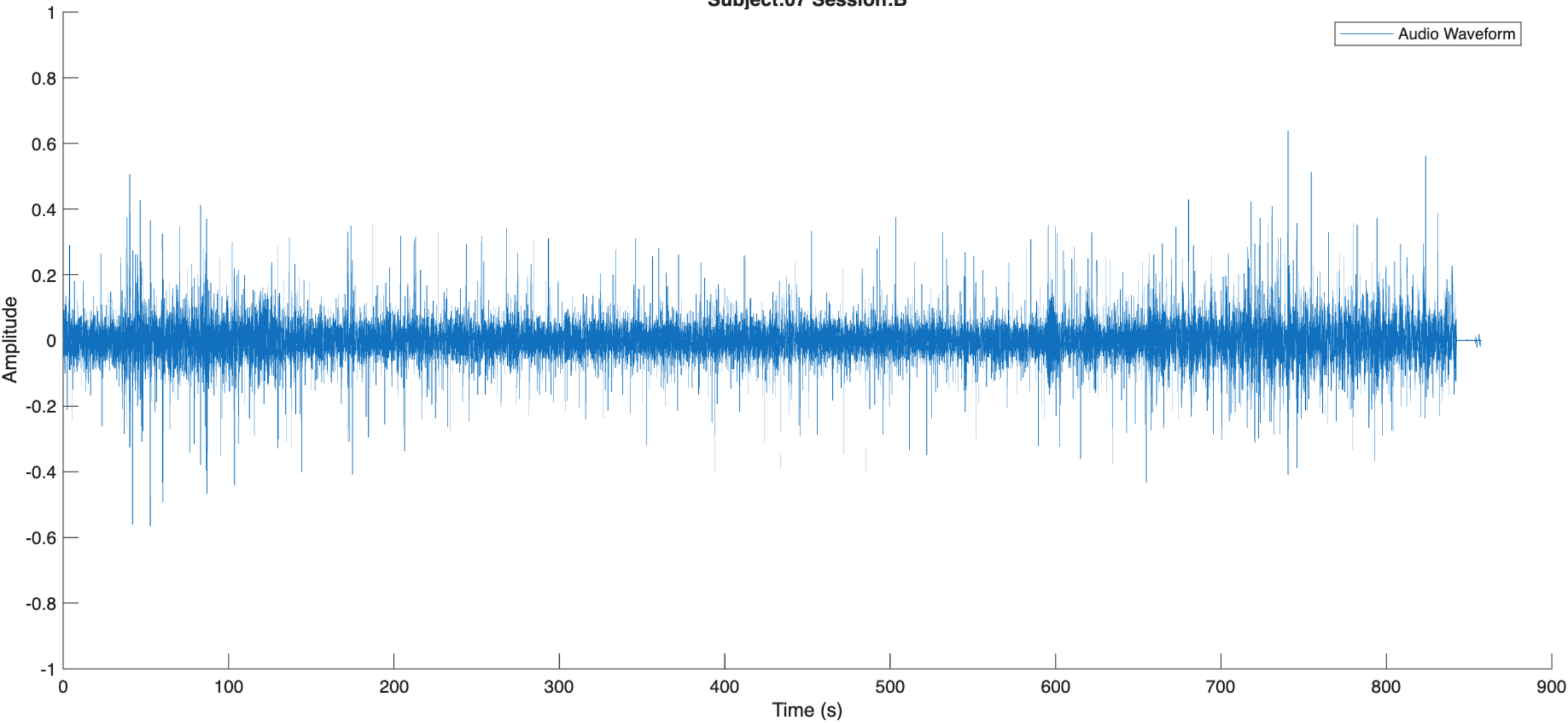
Subjects: 22 expert jazz musicians (various instrument specialties)

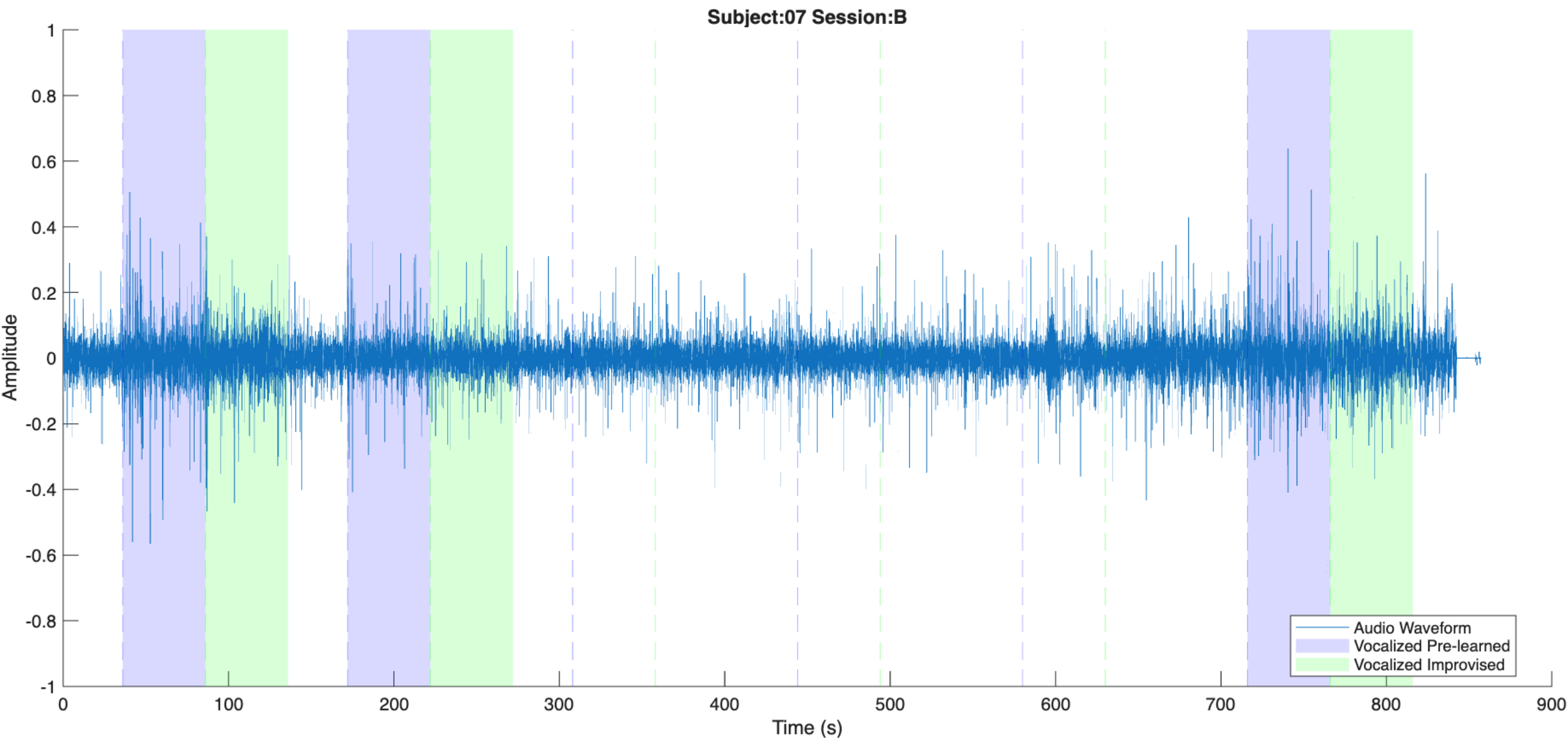
- **Sessions per subject:** 2
- **Tasks per session:** 6 (3 vocalized, 3 imagined)
 - Charlie Parker tunes: *Confirmation*, *Ornithology*, or *Anthropology*
- **Task structure:**
 - Sing/imagine tune (3 sec)
 - Metronome clicks at 154 BPM (3 sec)
 - Melody (50 sec)
 - Improvisation (50 sec)
 - Self-assessment questions (30 sec)

Full session ~14 min



Subject:07 Session:B



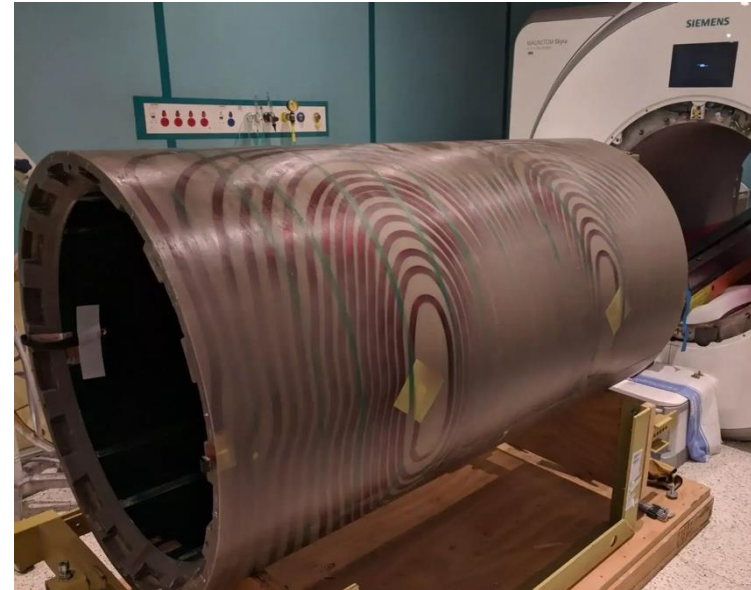


Cold Head



© CryoSrv / cryosrv.com

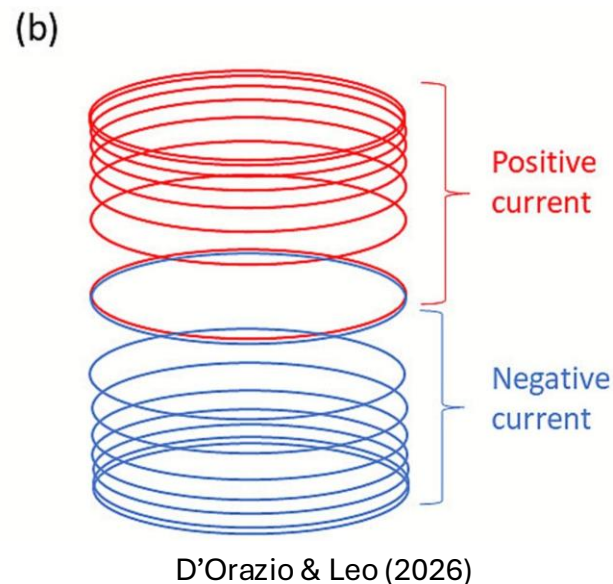
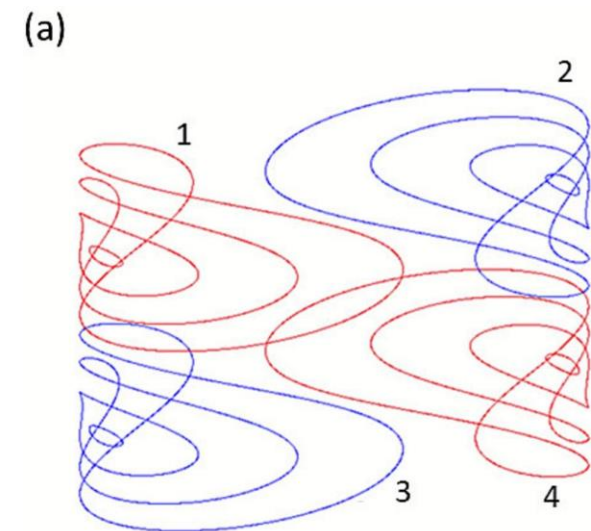
Gradient Coils



© DirectMed Imaging / directmedimaging.com

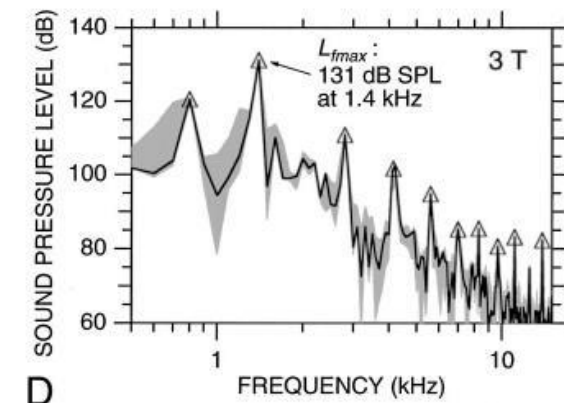
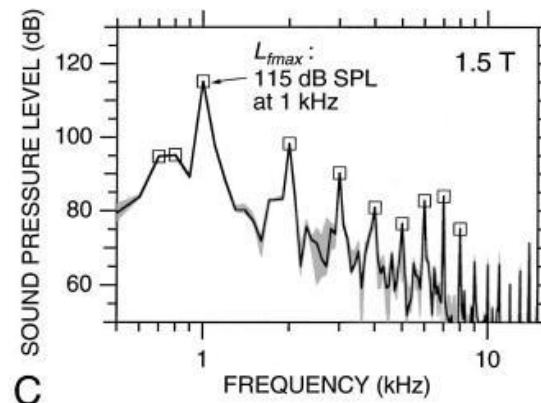
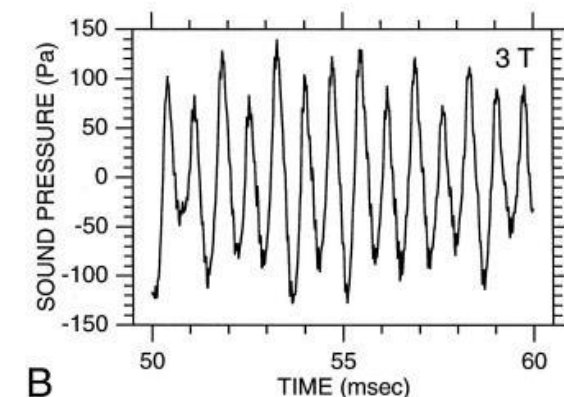
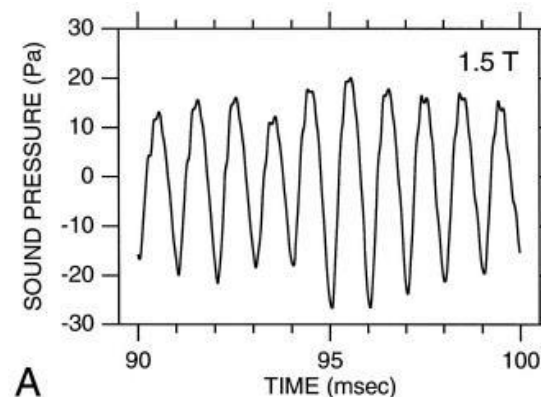
IT'S LOUD IN THERE!

Gradient Coils are Noisy!

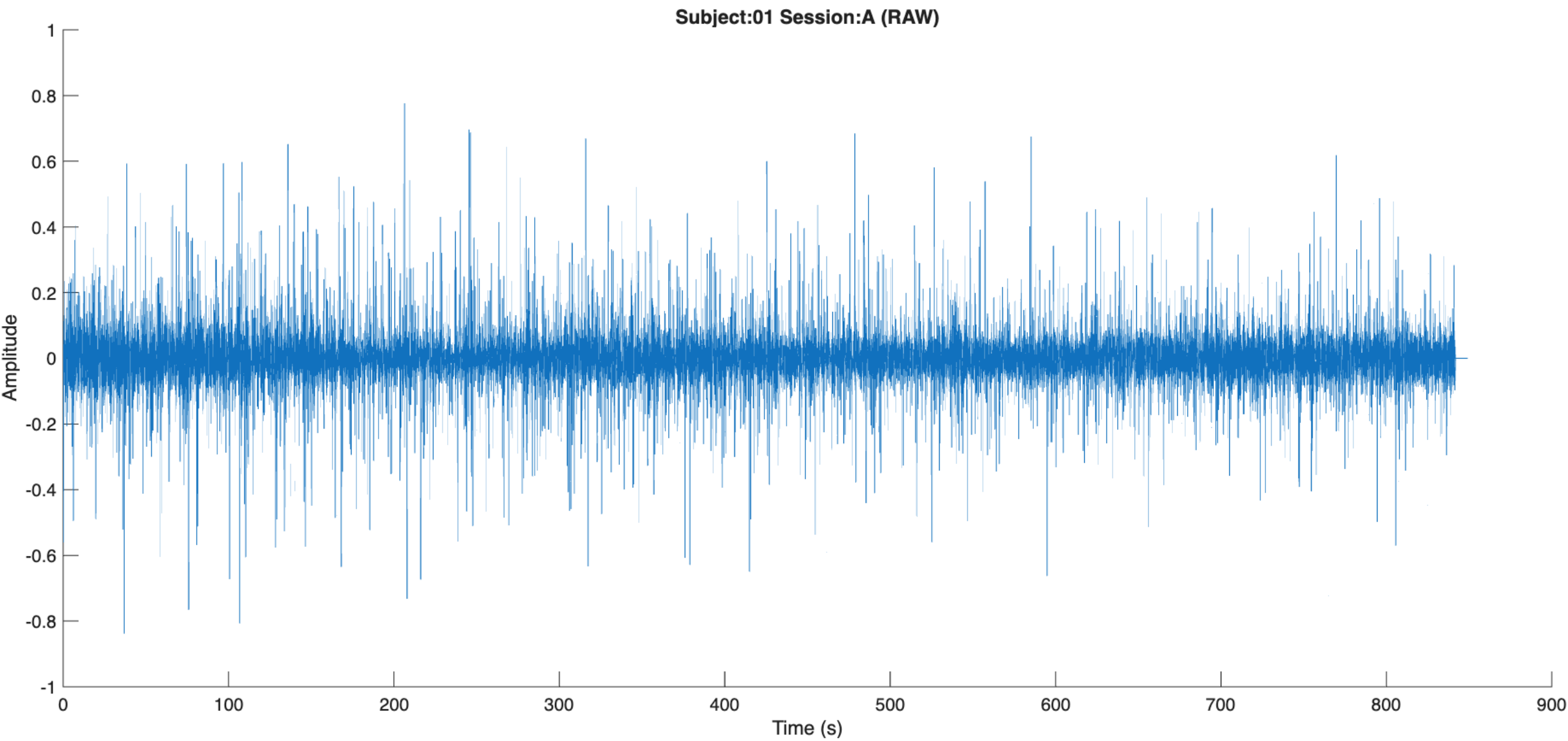


Lorentz Force Law

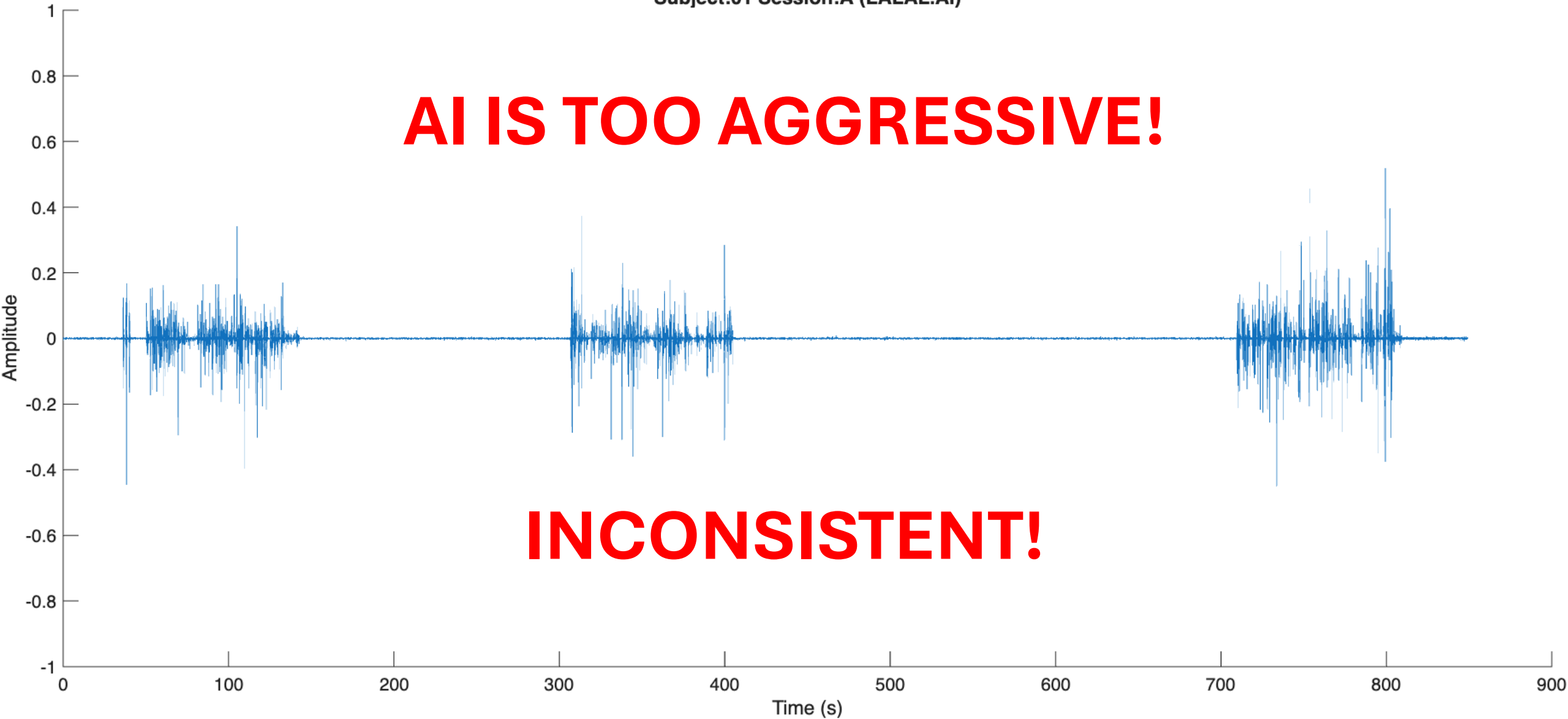
$$F = IL \times B$$



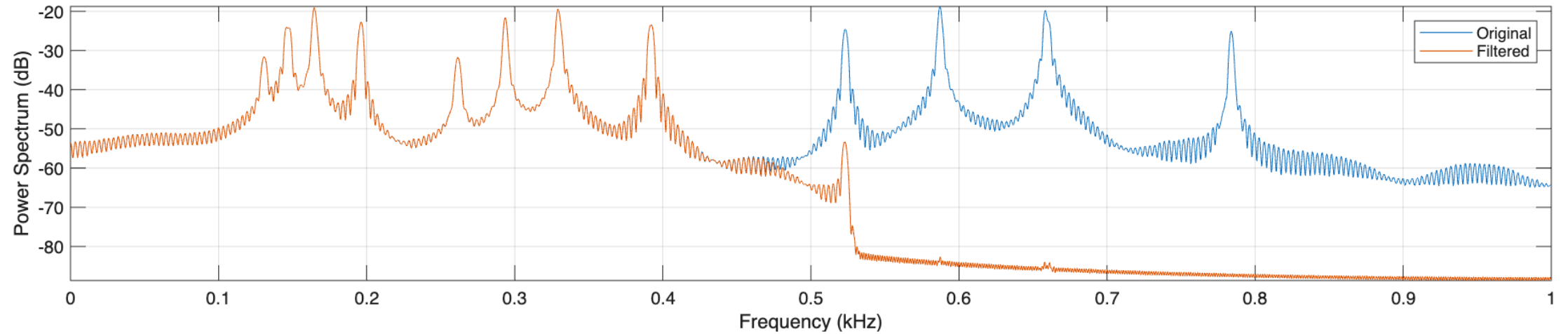
Ravicz et al., (2000)



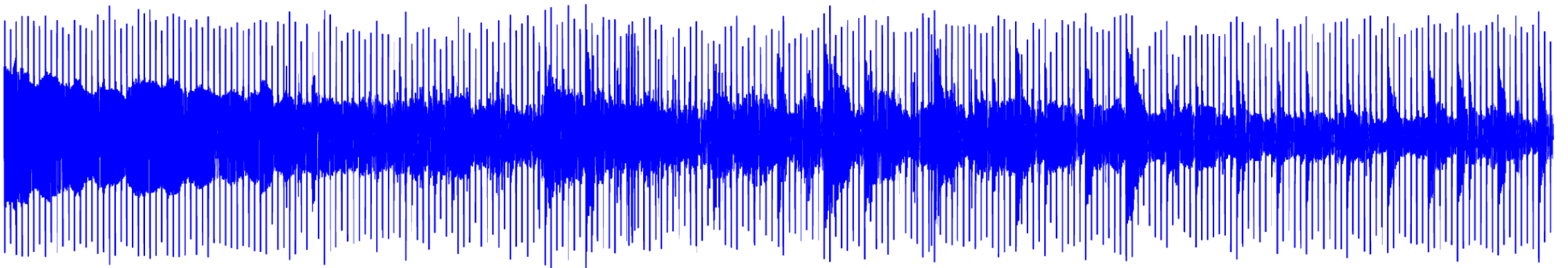
Subject:01 Session:A (LALAL.AI)



LOW PASS FILTER?



SPECTRAL SUBTRACTION?

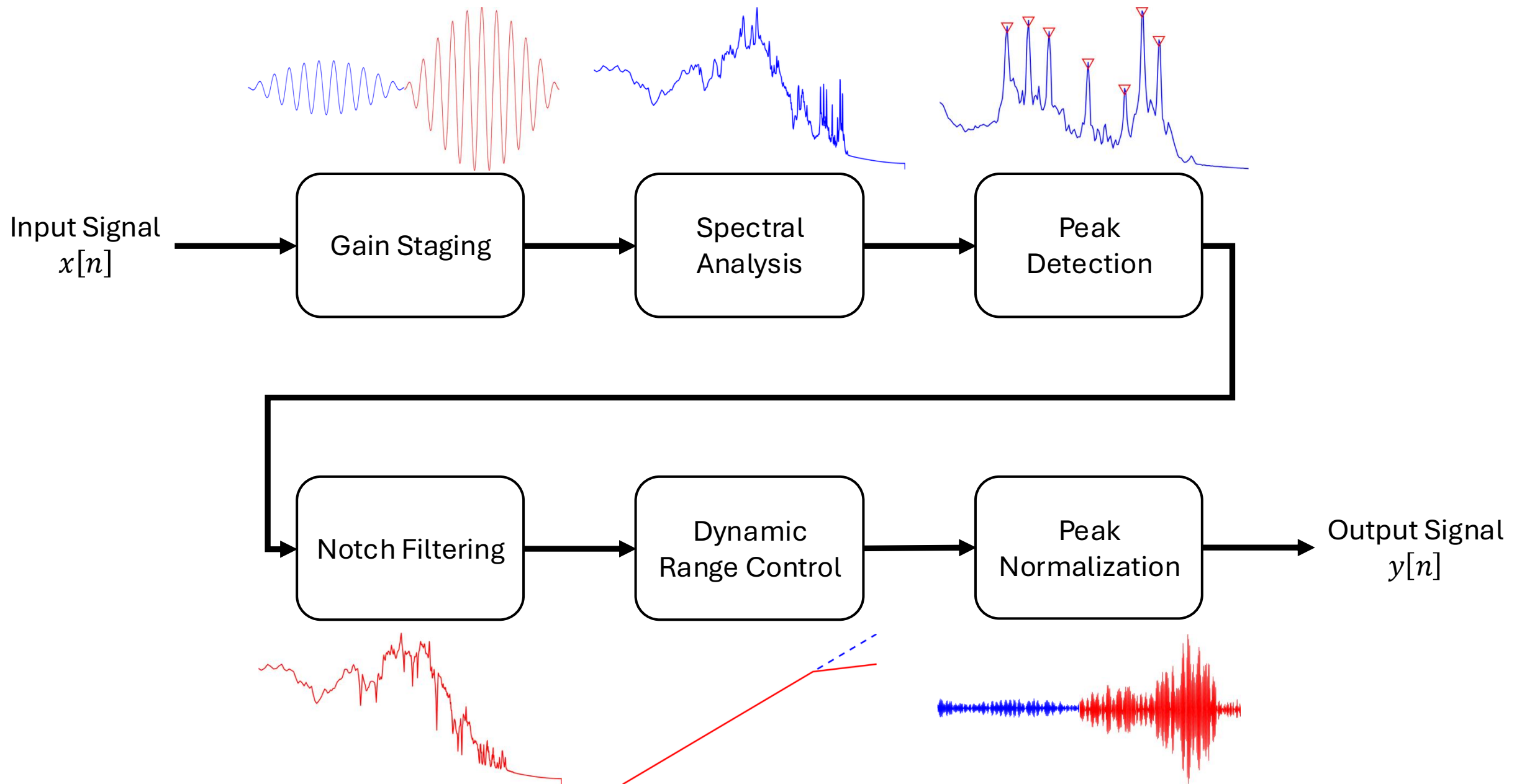


S.P.E.A.R.

Spectral Peak Elimination &

Amplitude Regulation

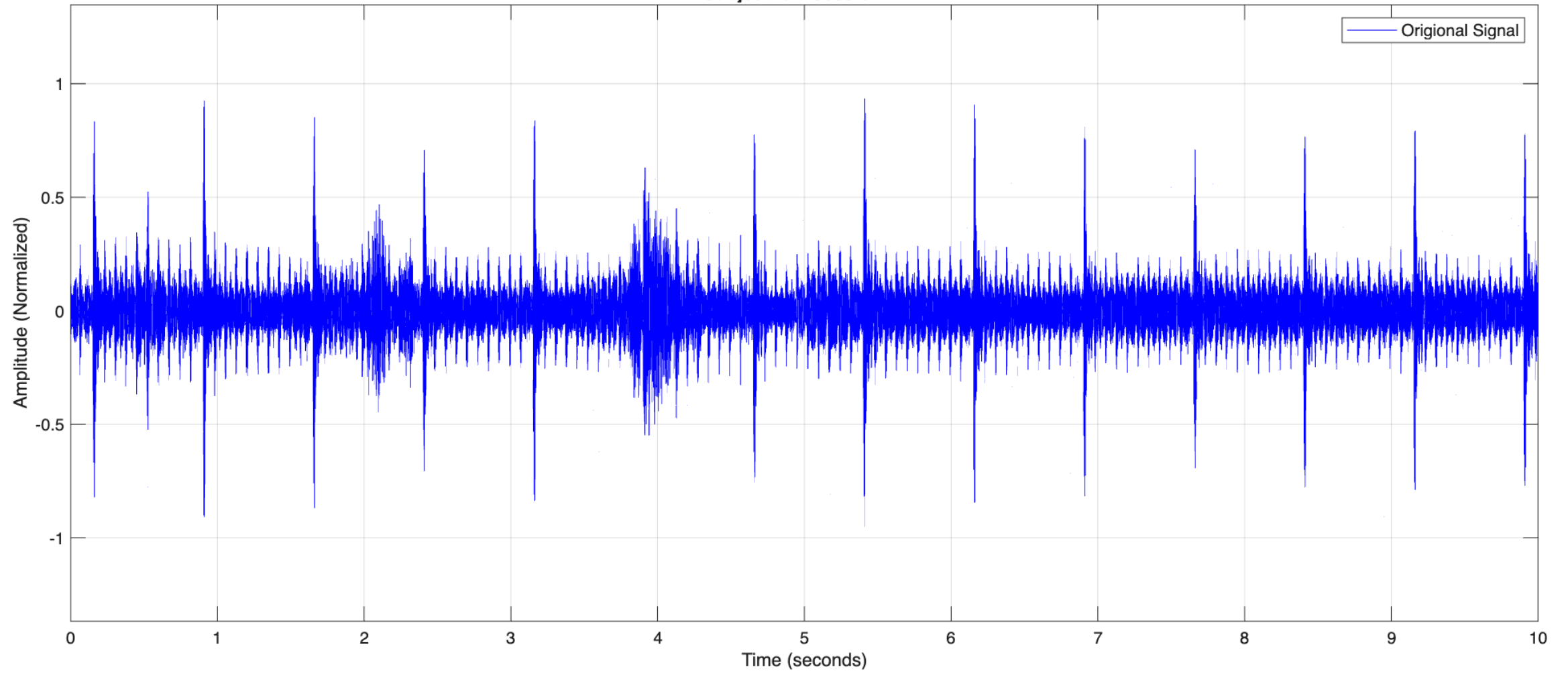
A MATLAB-based digital signal processing tool for removal of acoustic noise from fMRI-acquired jazz scat singing



Signal Processing = Sculpting

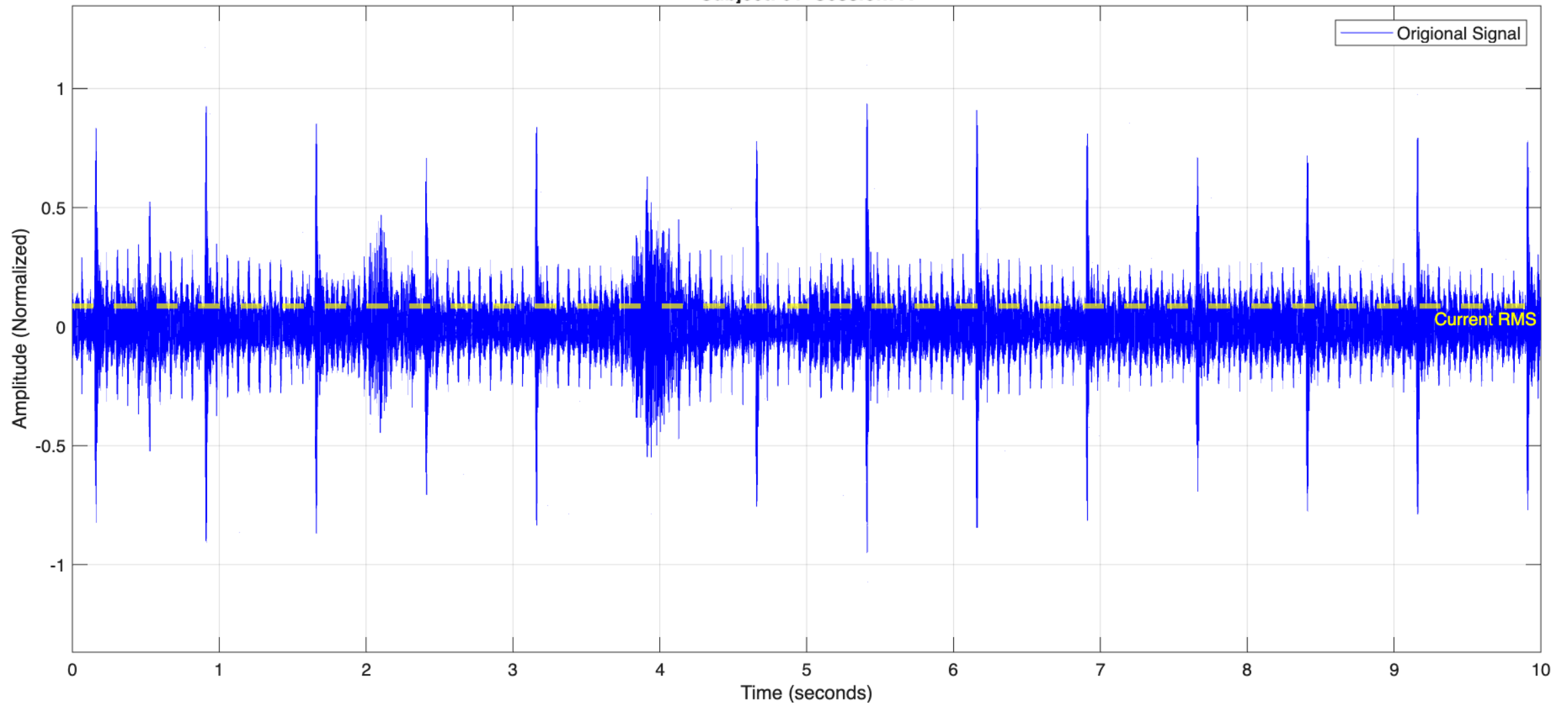


Subject: 01 Session: A

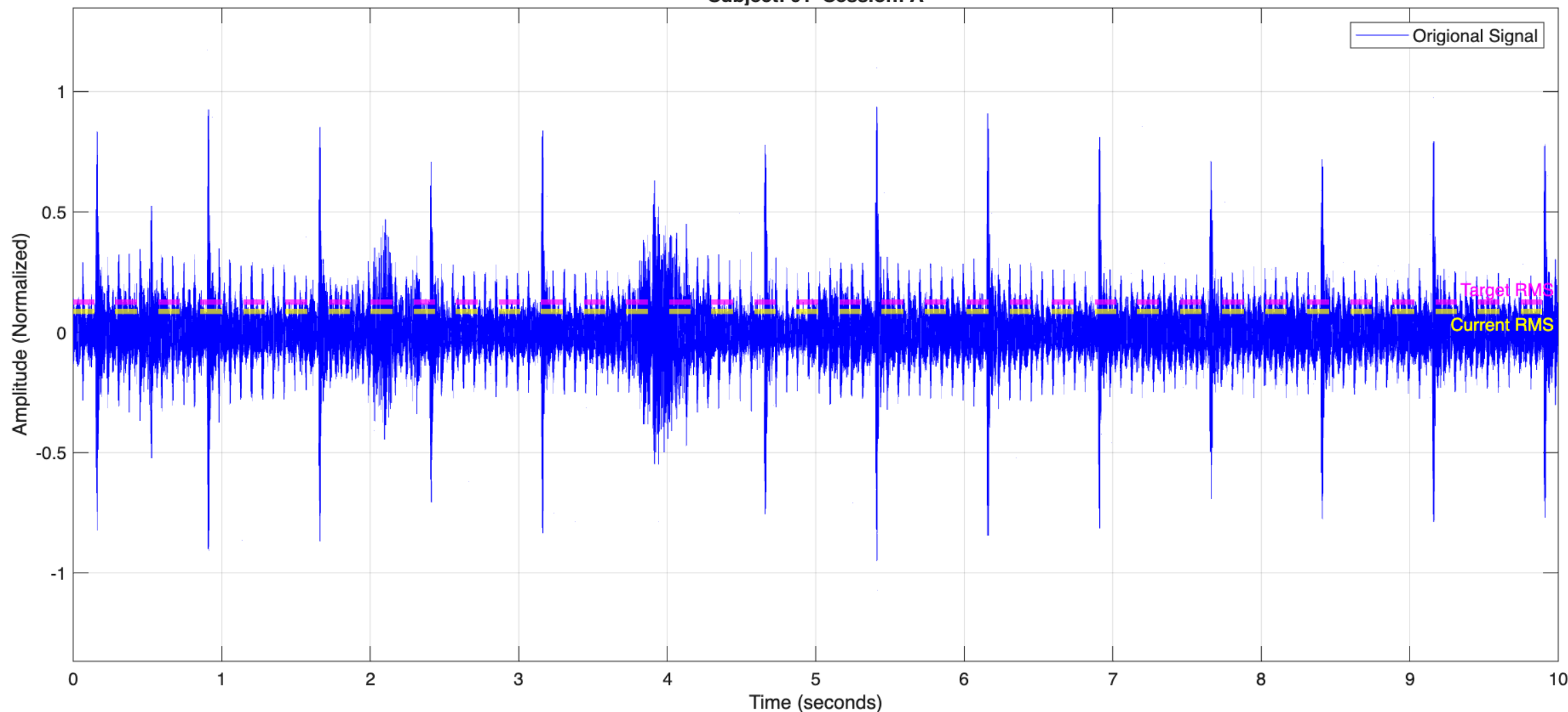


Gain Staging: Root Mean Square (RMS) Scaling

Subject: 01 Session: A

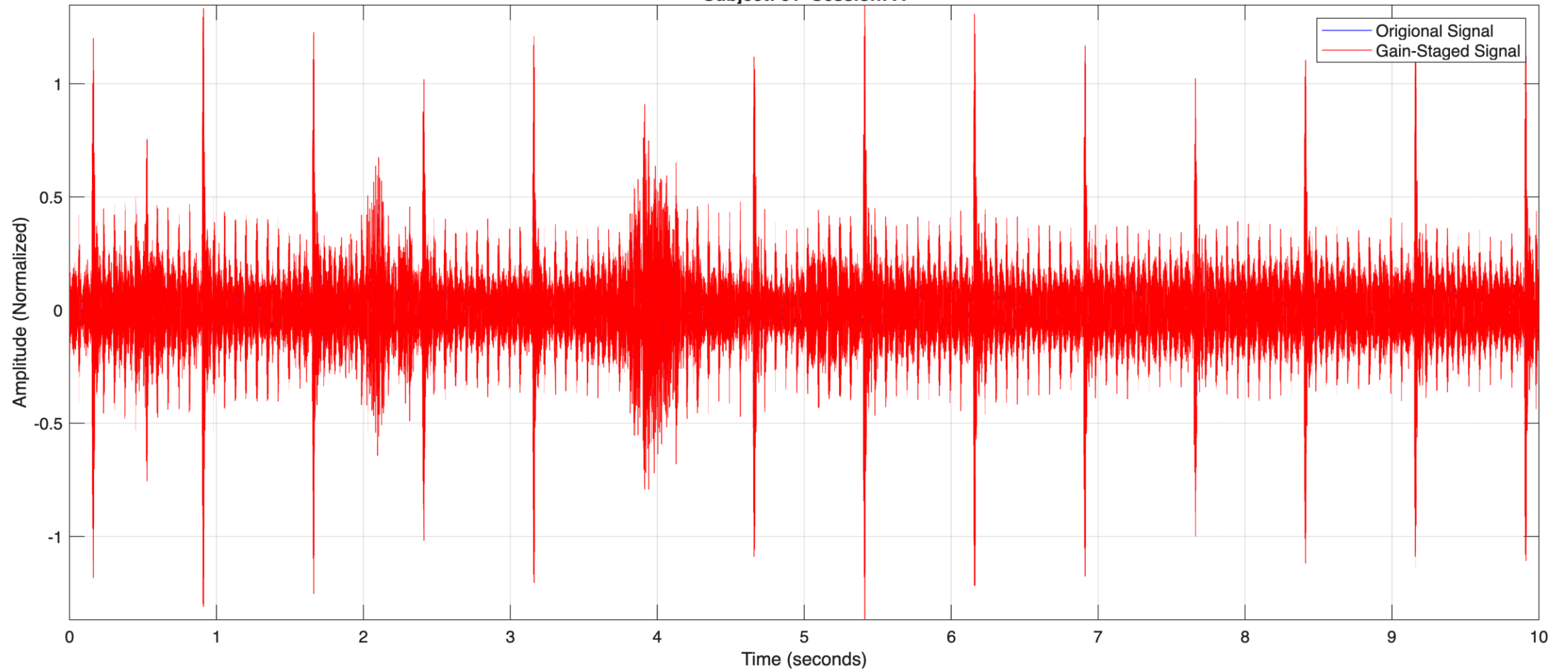


Subject: 01 Session: A

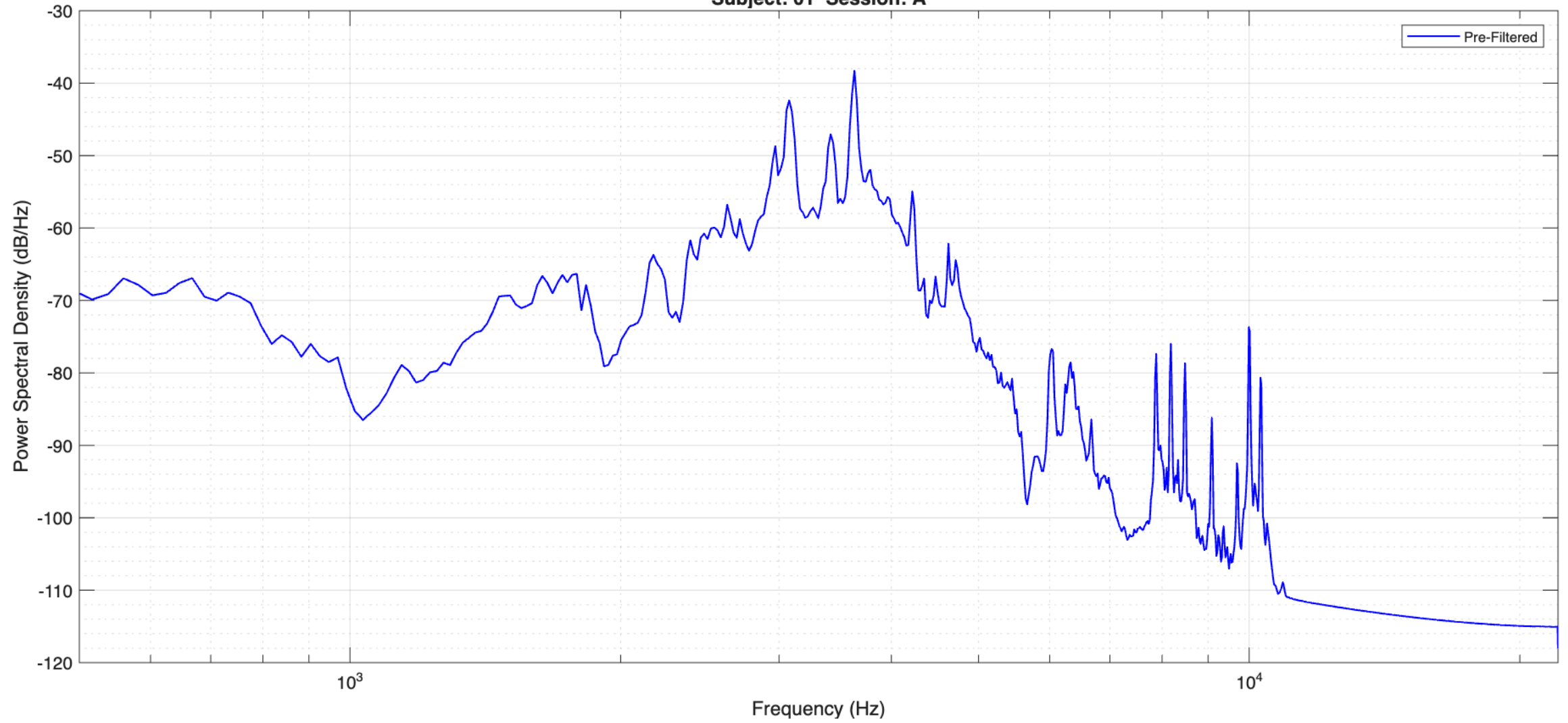


CONSISTENT RMS ACROSS SESSIONS!

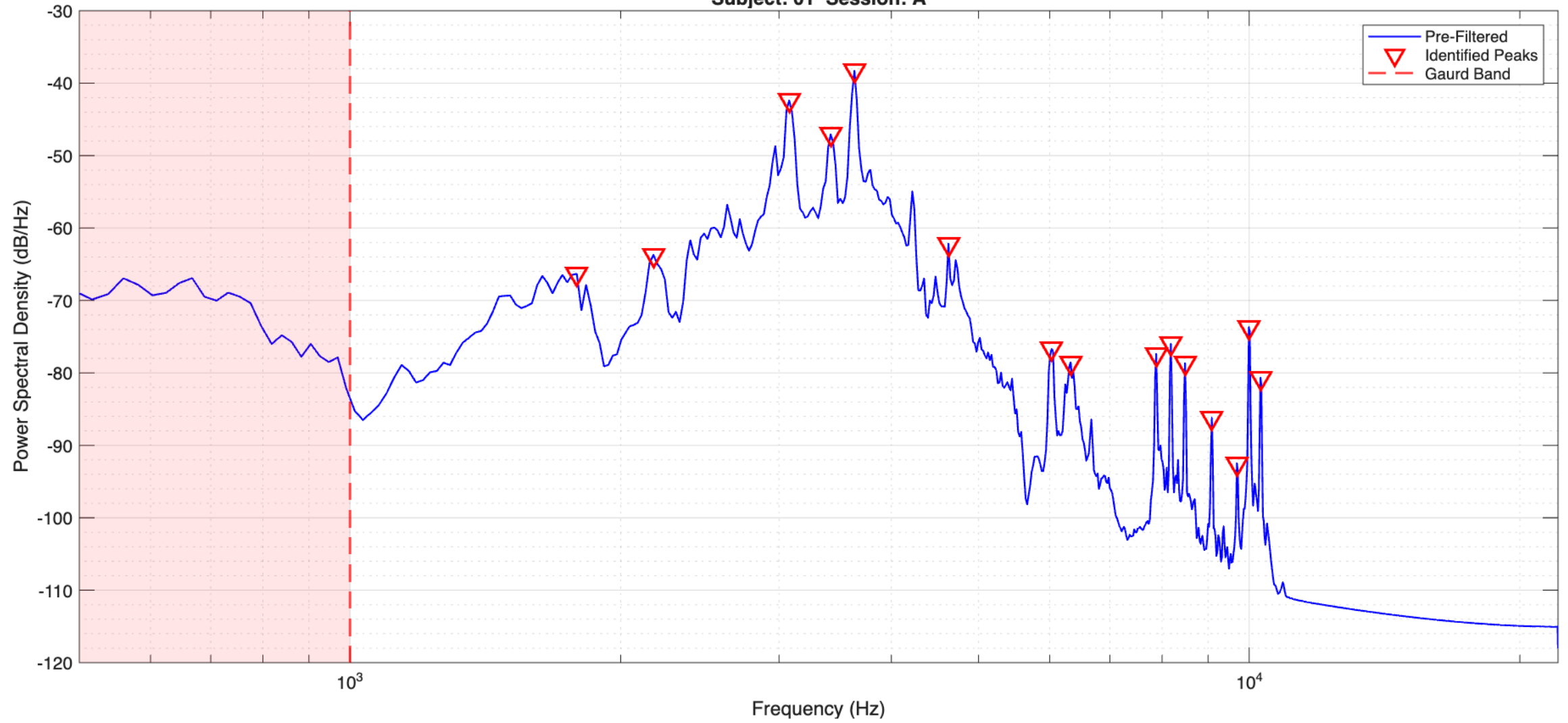
Subject: 01 Session: A



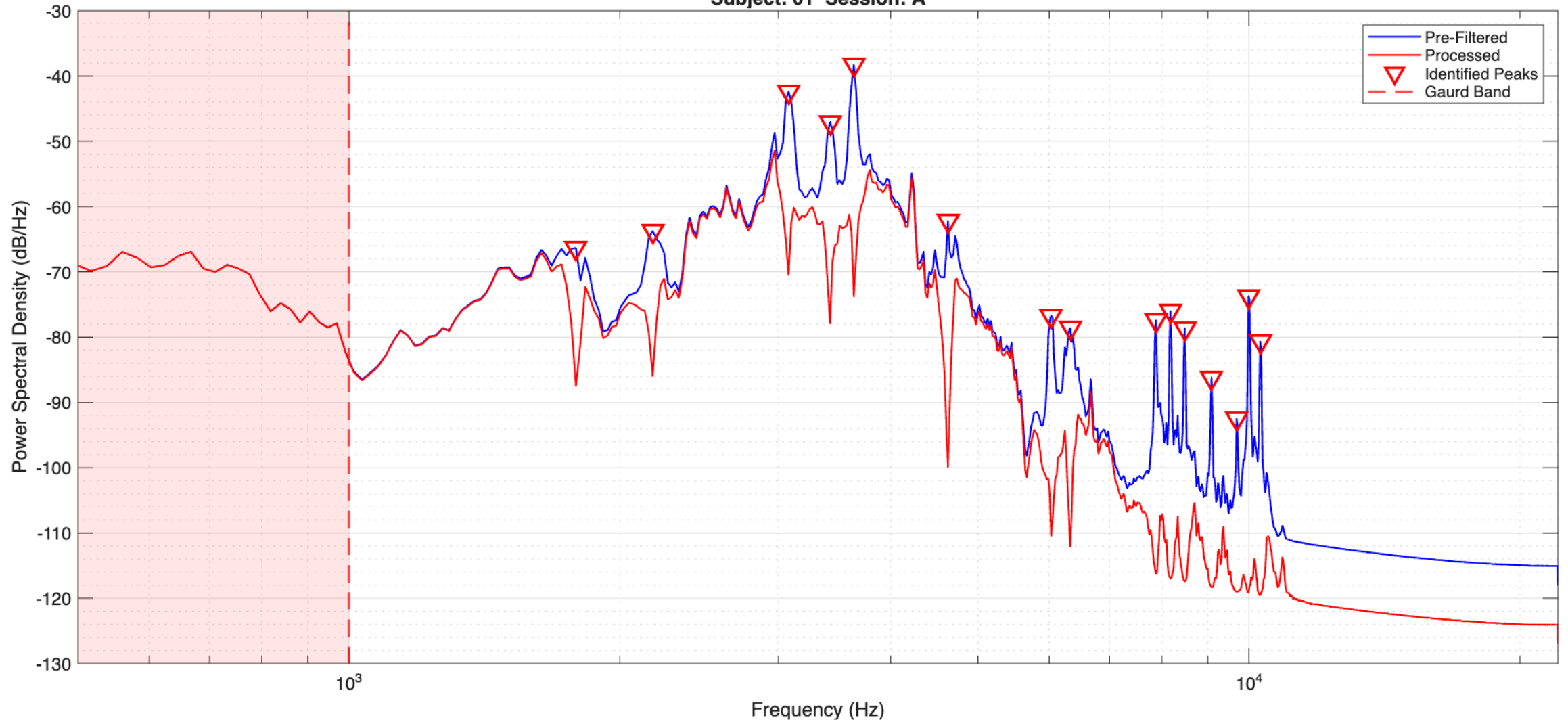
Subject: 01 Session: A



Subject: 01 Session: A



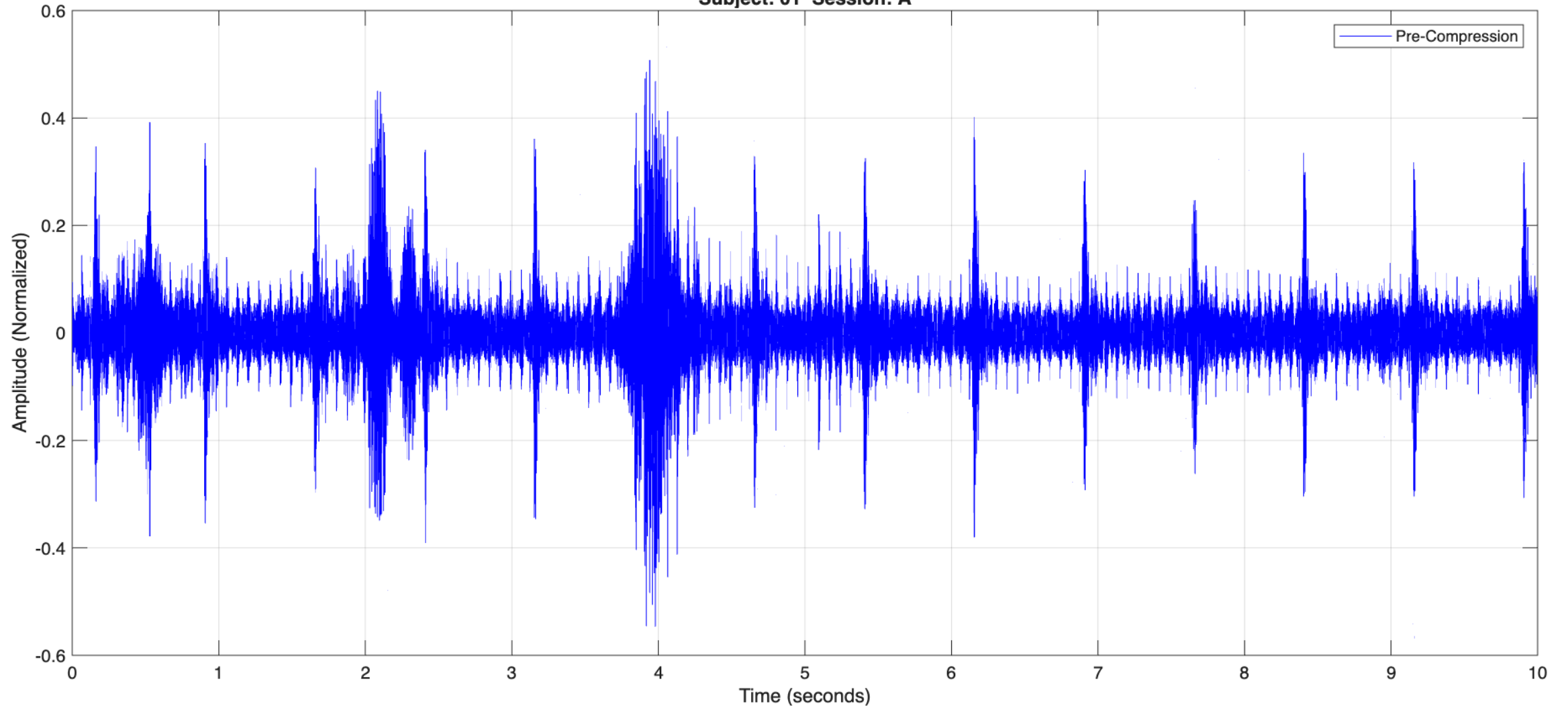
Subject: 01 Session: A



PULL THOSE PEAKS DOWN!

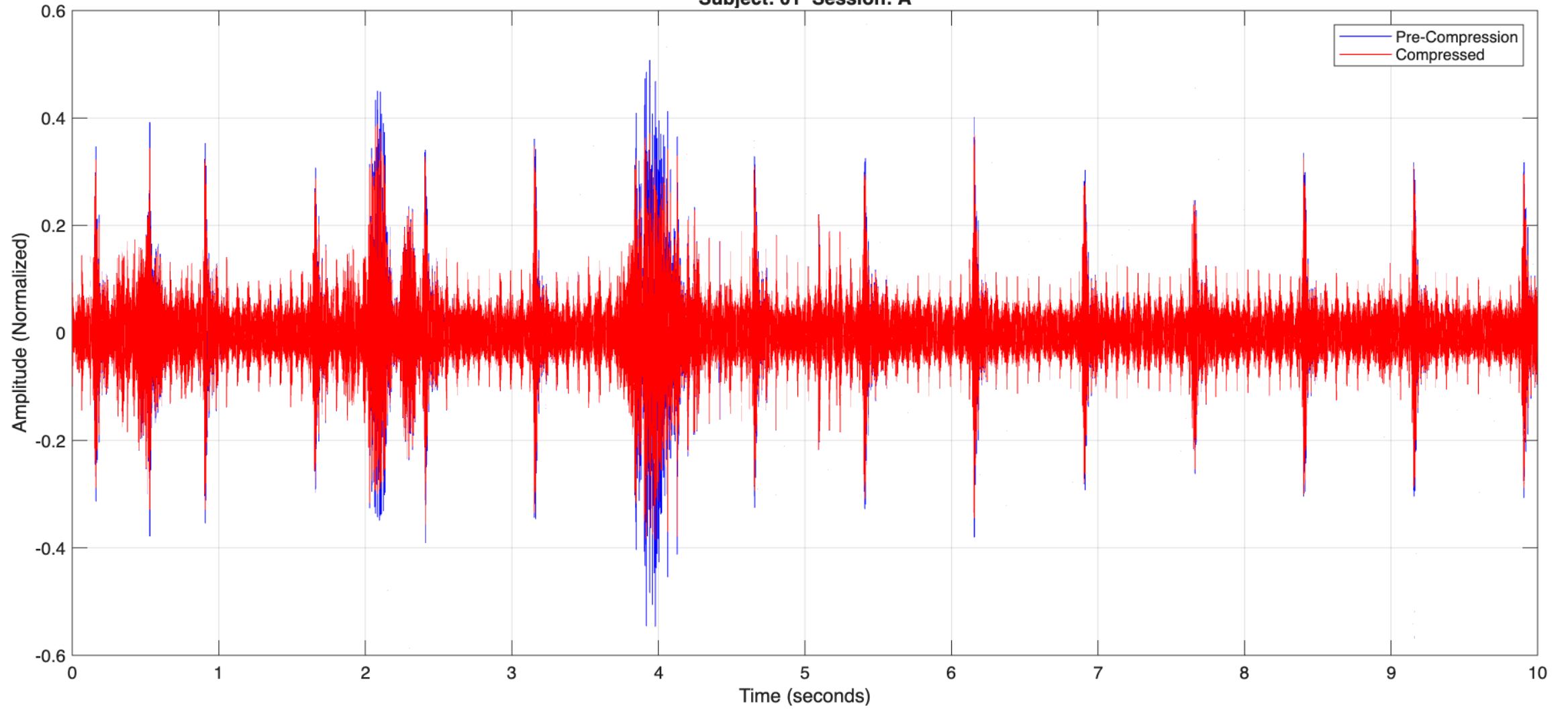
Dynamic Range Control: Compression

Subject: 01 Session: A



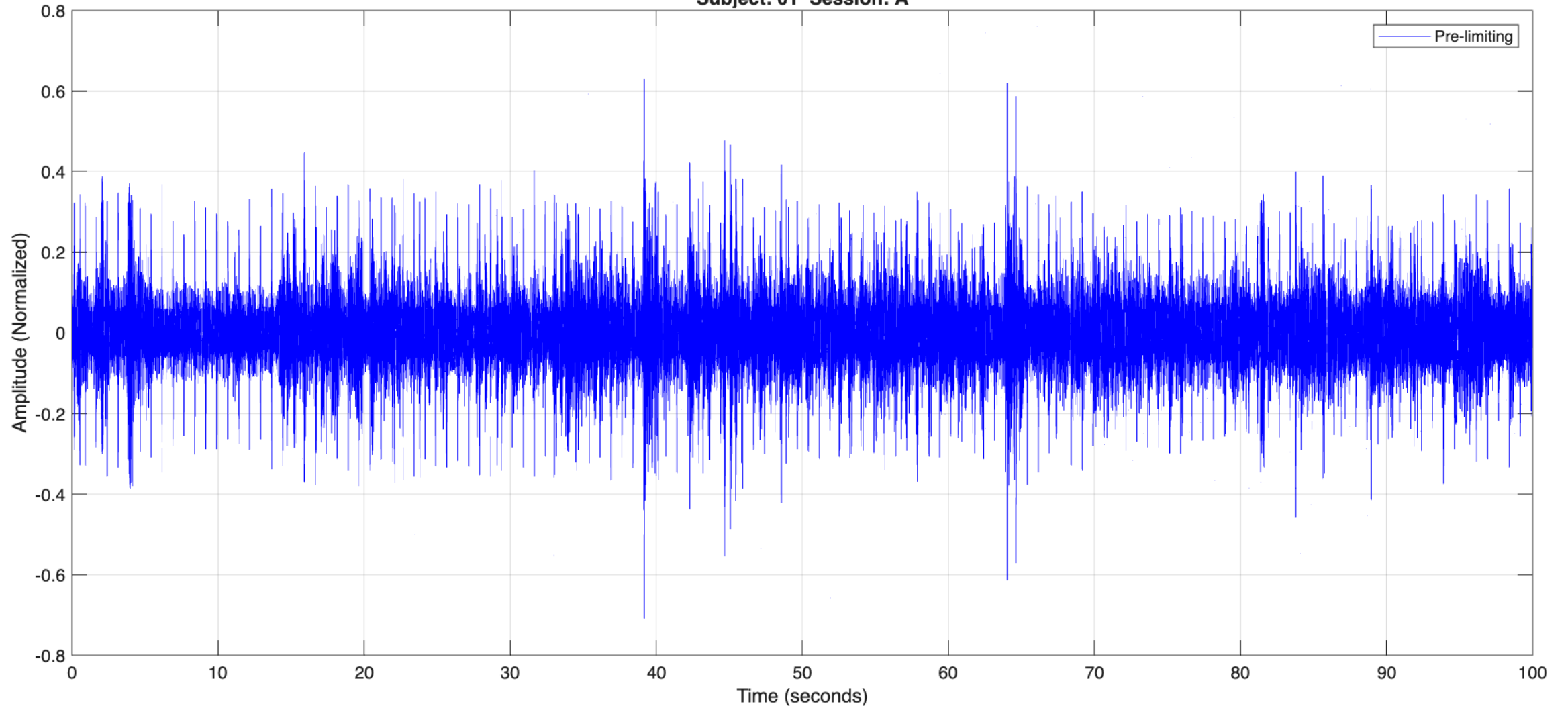
Dynamic Range Control: Compression

Subject: 01 Session: A



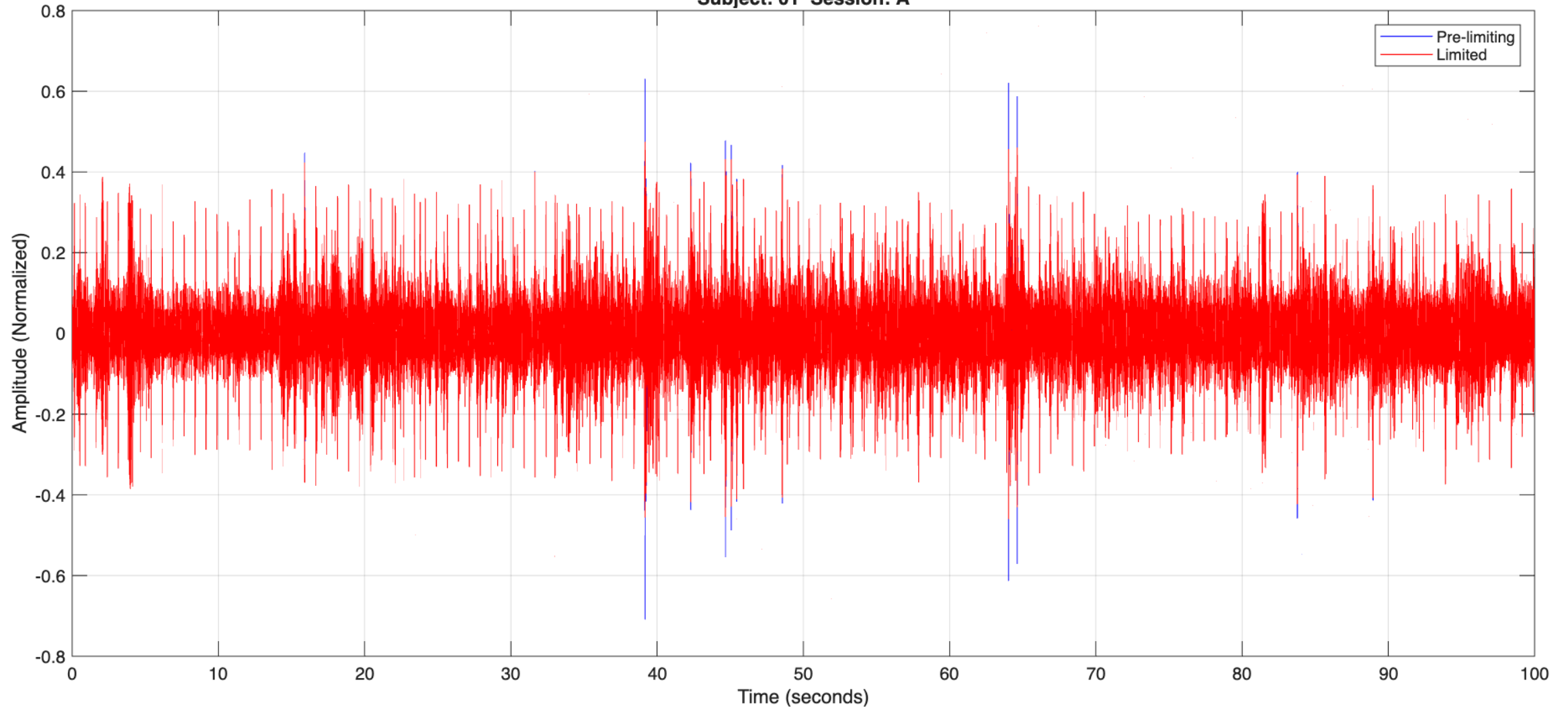
Dynamic Range Control: Limiting

Subject: 01 Session: A



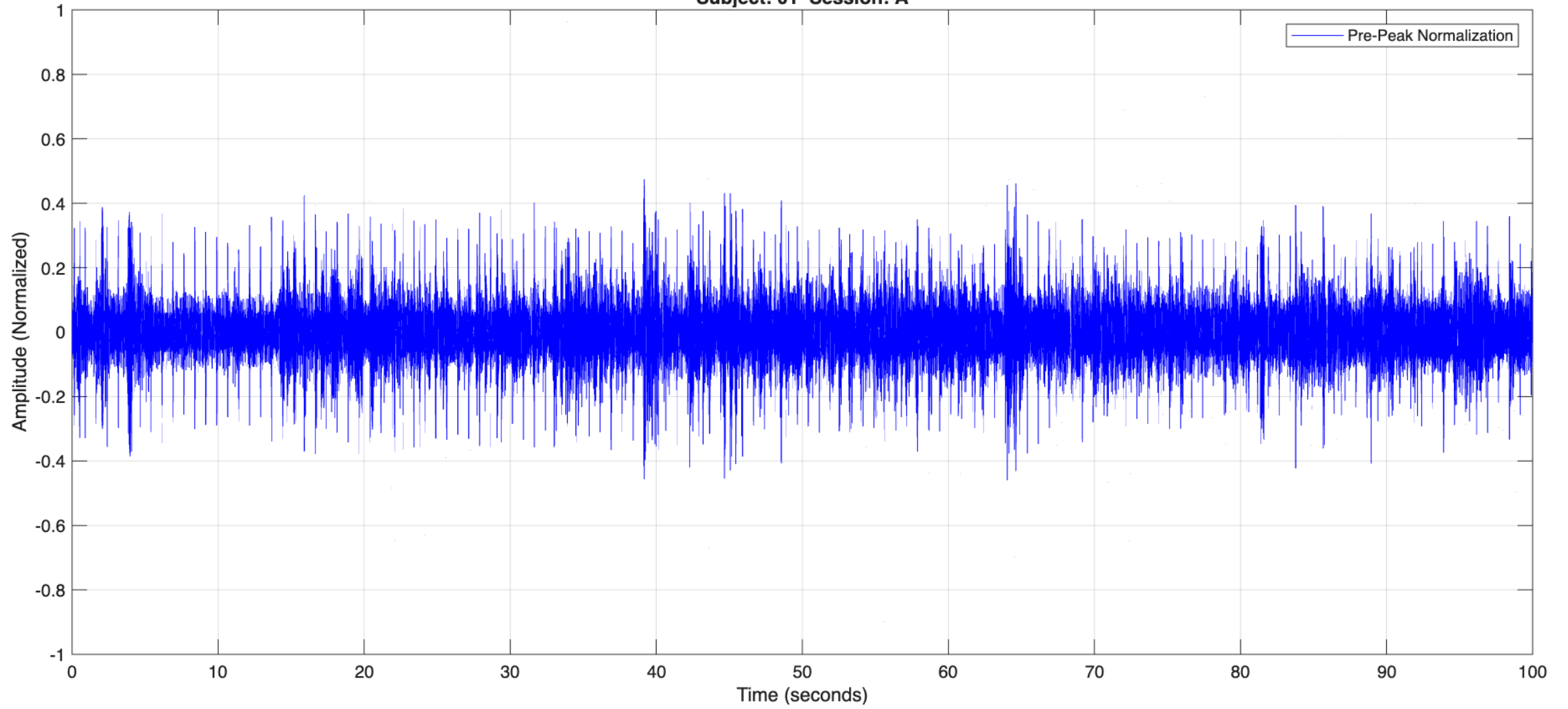
Dynamic Range Control: Limiting

Subject: 01 Session: A

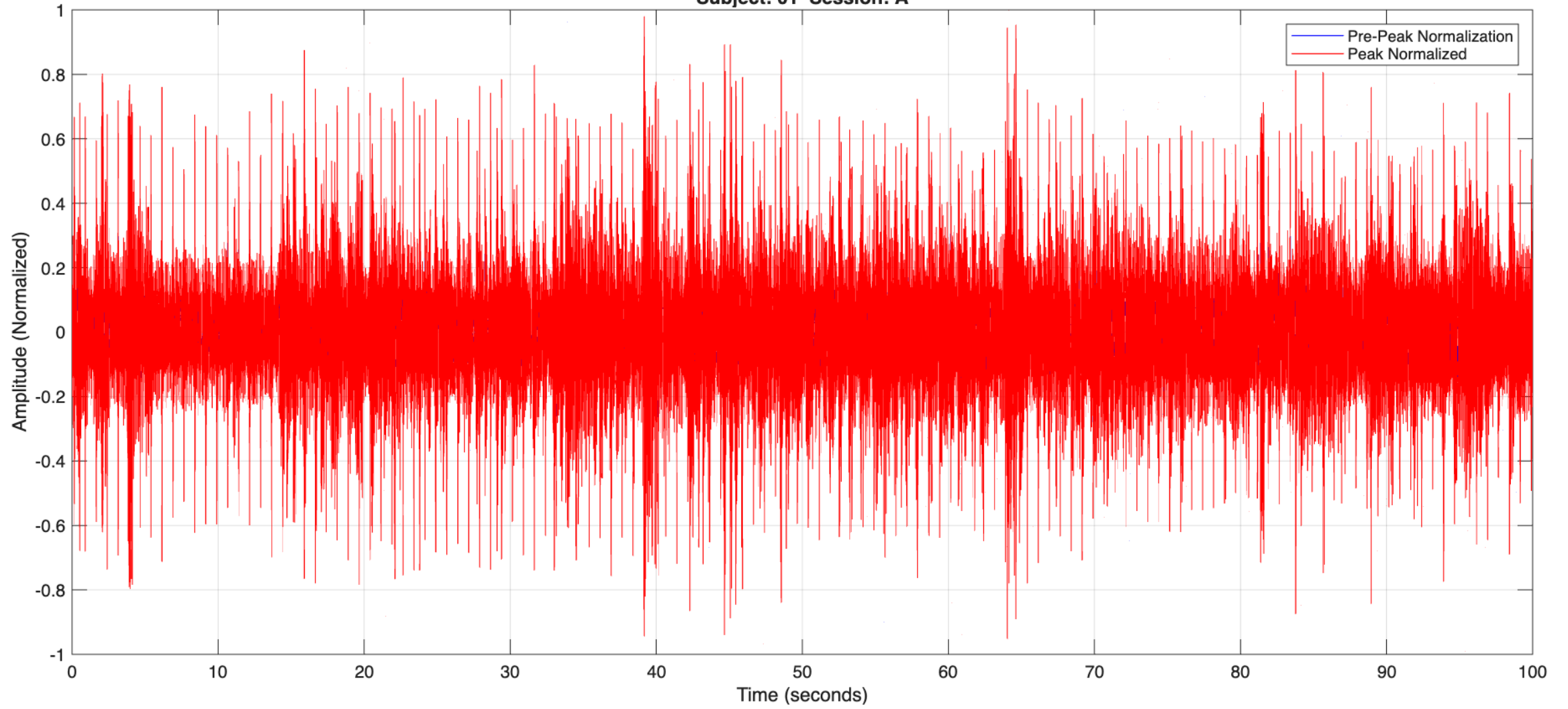


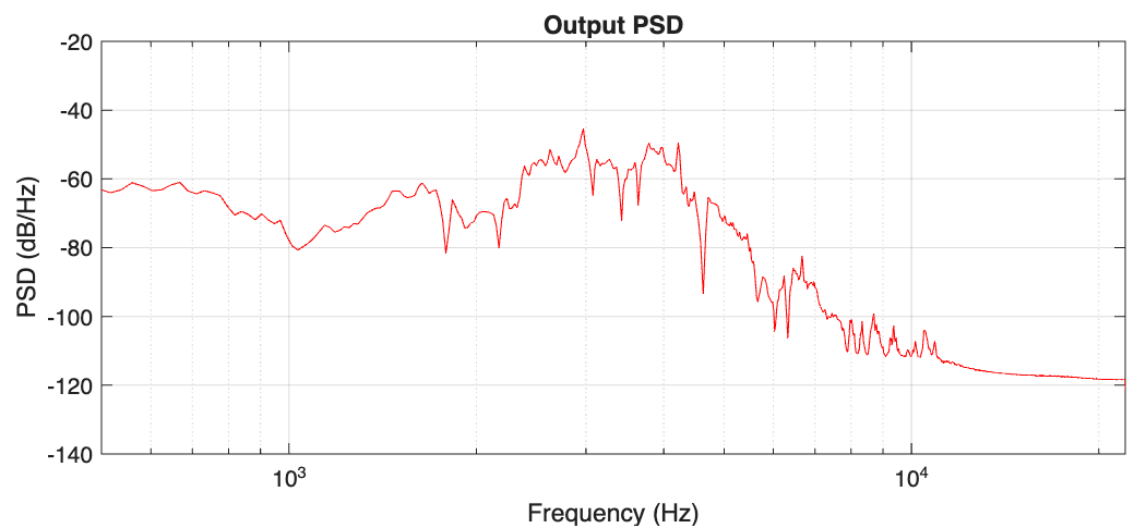
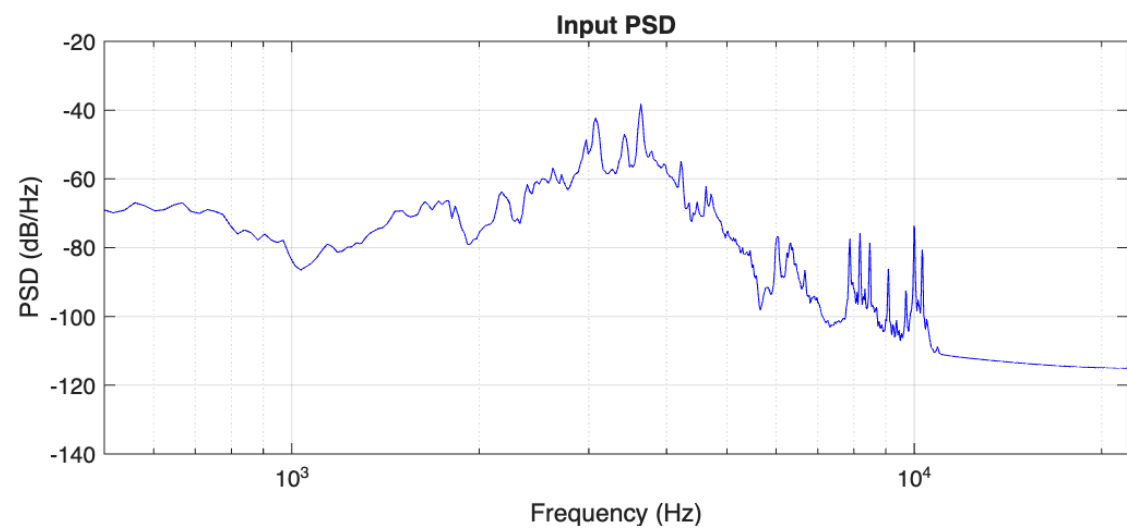
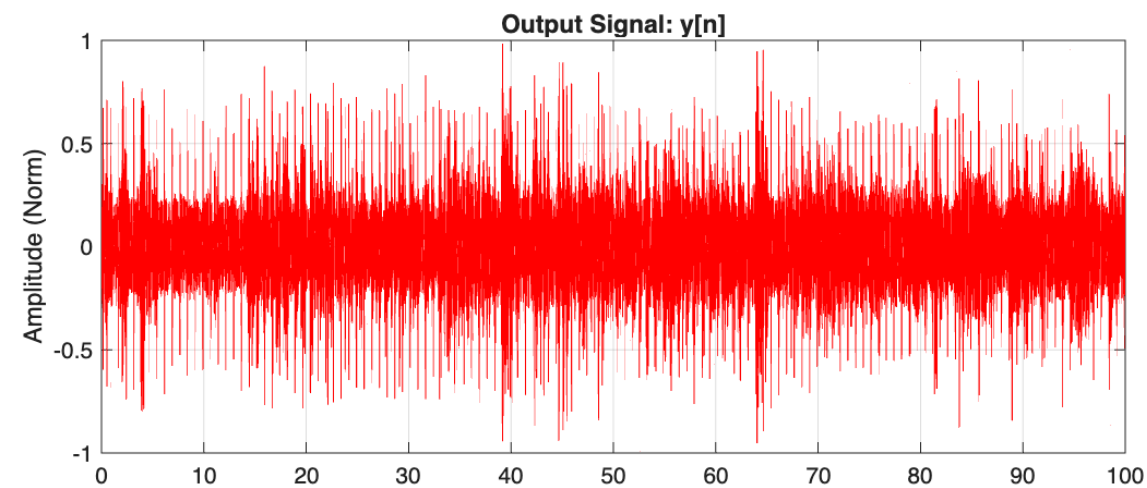
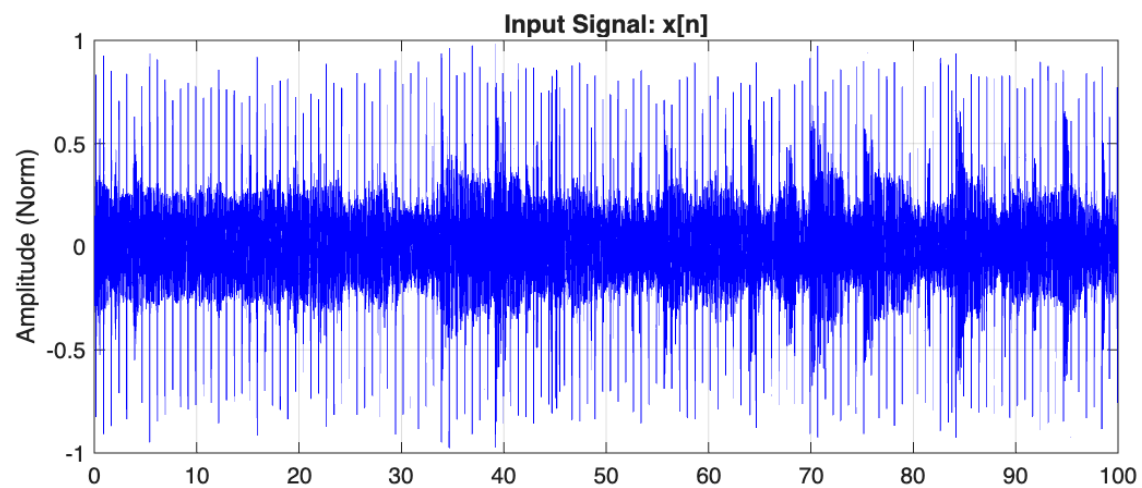
GET SOME HEADROOM!

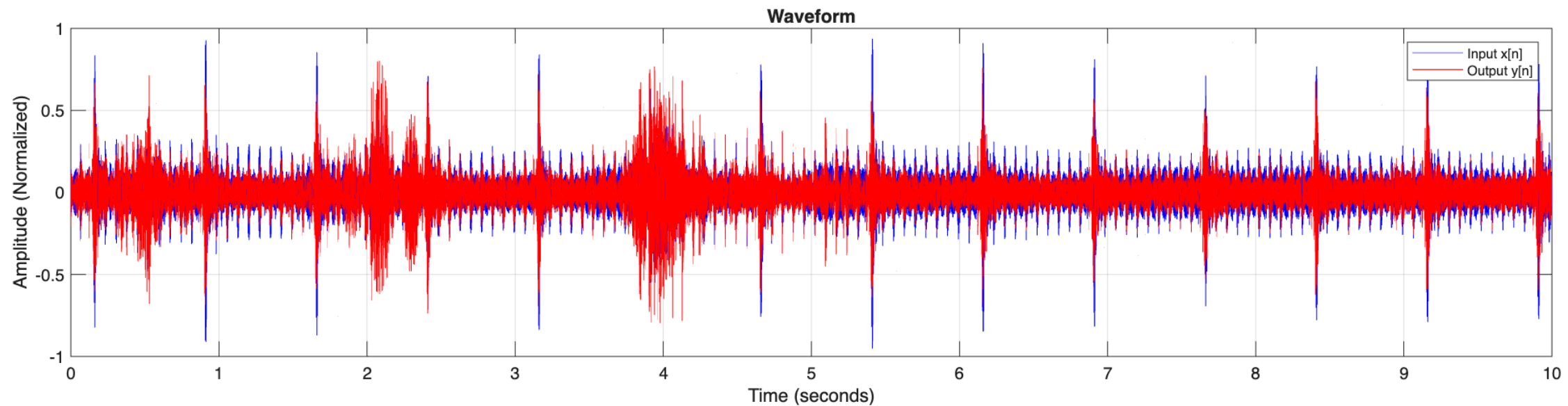
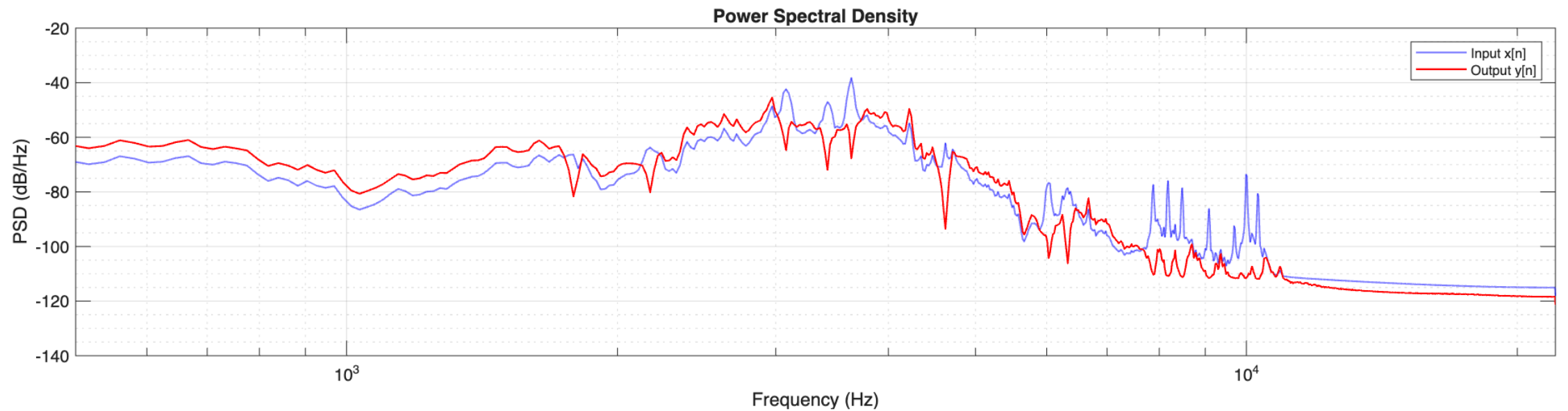
Subject: 01 Session: A



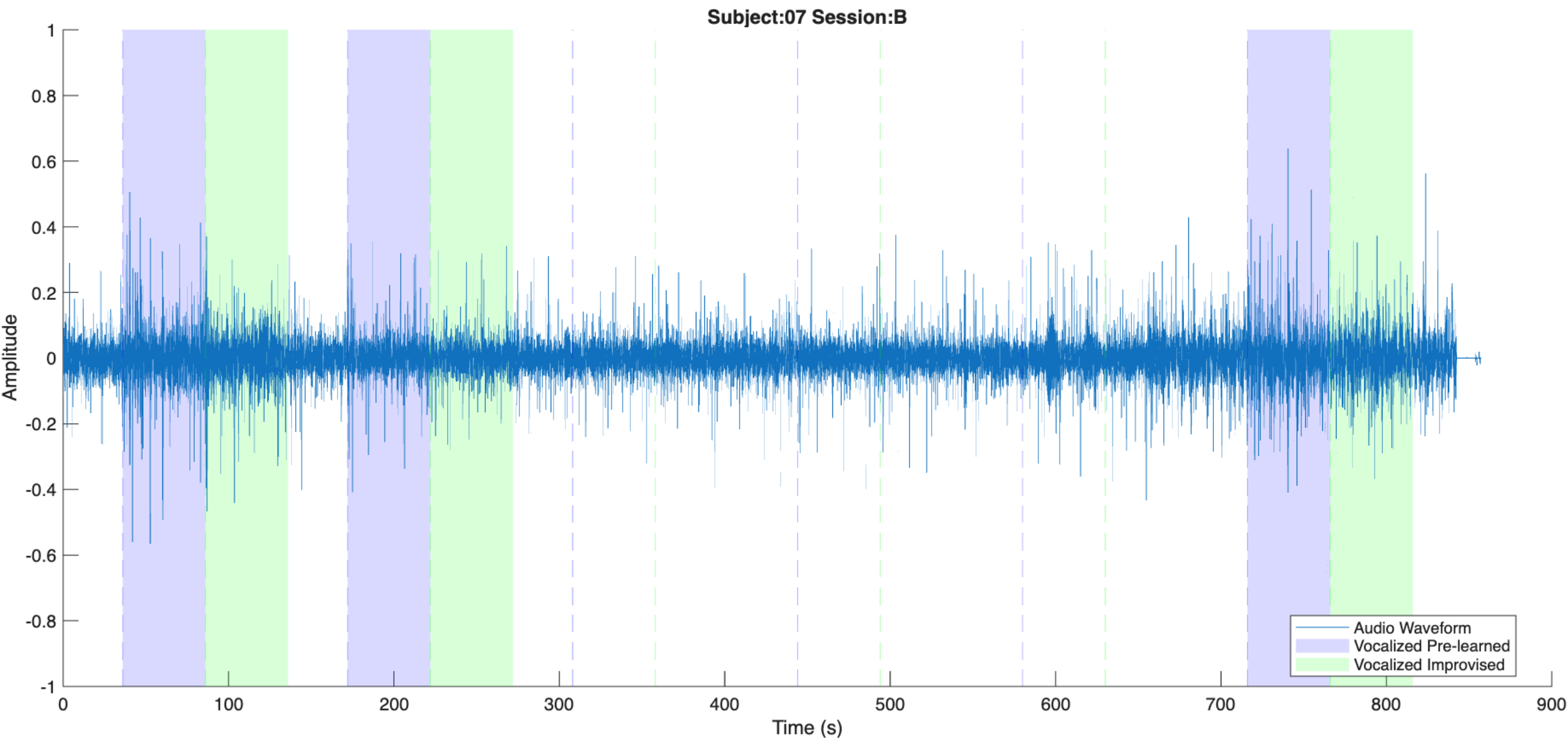
Subject: 01 Session: A

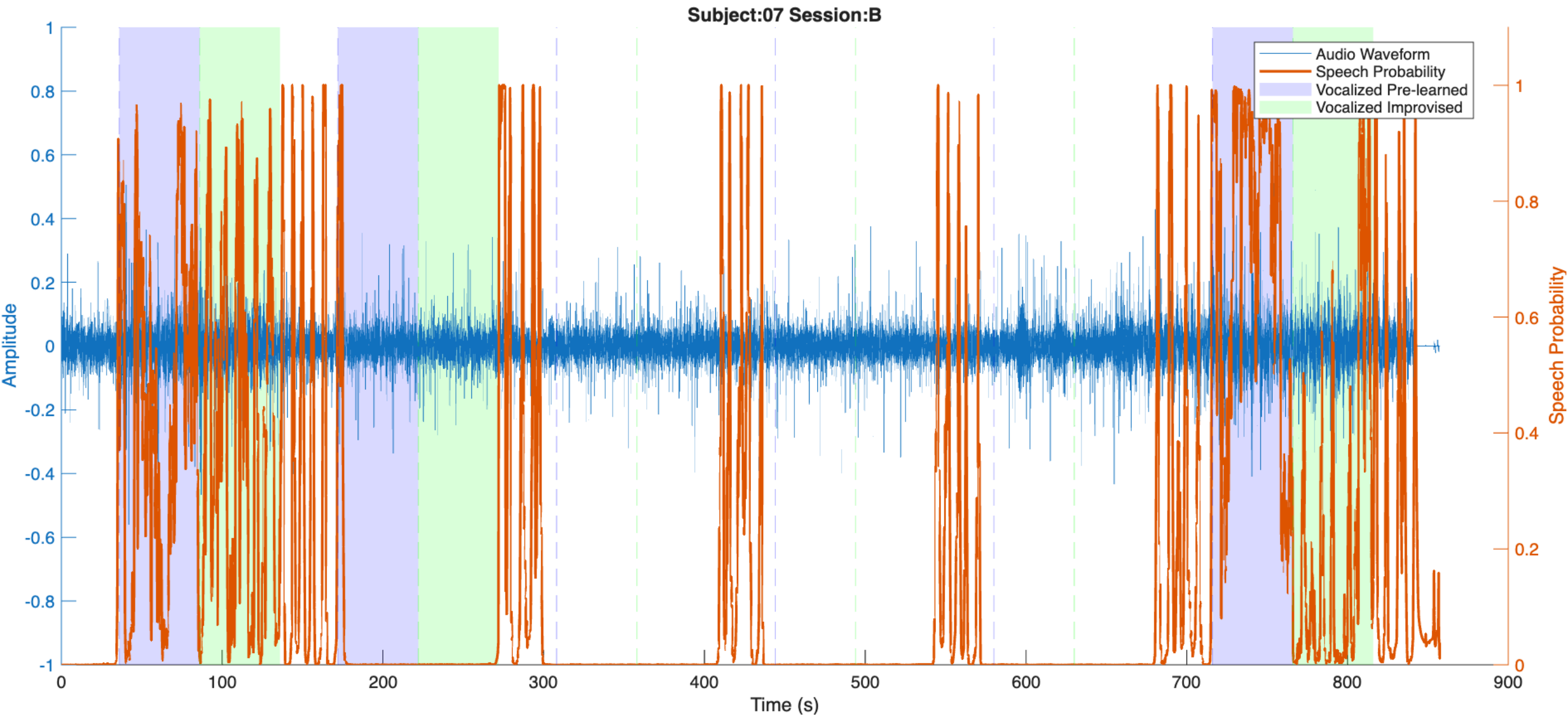
**MAKE IT LOUDER!**





LET'S LISTEN!





MY JOB: QUALITY CONTROL

Where We Stand

- S.P.E.A.R. improves vocal intelligibility & signal-to-noise ratio
- Adaptive filtering approach is generalizable for many pursuits
- Experts are currently rating solos via our Qualtrics interface

Looking Forward

- Data driven parameter selection for compressor and limiter – windowed S.P.E.A.R.
- Correlation of scores w/ data

12:29

Georgia State University

▶ 0:00 / 1:05

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How often does the soloist imply

12:29

How often does the soloist imply the underlying chord structure?

Chord

1 (Never)	☐
2 (Rarely)	☐
3 (Sometimes)	☐
4 (Often)	☐
5 (Always)	☐

How often does the soloist use interesting melodic ideas?

Melody

References & Image Credits

Research Articles

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Media & Visual Sources

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